# Physics

## 1. MECHANICS

1. The distance between a town and a mill is 30 km. A man started to walk from the mill to the town at 6:30 a.m., while a cyclist left the town for the mill at 6:40 a.m., riding at a speed of 18 km/h. The man met the cyclist after walking 6 km. Determine at what time they met and the man's speed?

   Also find the place where the man was when he met the twelfth bus coming from the town and the number of buses which overtook the cyclist if bus traffic begins at 6 a.m. The buses leave their terminals every 15 minutes and their speed is 45 km/h.

2. Two tourists who are at a distance of 40 km from their camp must reach it together in the shortest possible time. They have one bicycle which they decide to use in turn. One of them started walking at a speed of $v_1$ = 5 km/h and the other rode off on the bicycle at a speed of $v_2$ = 15 km/h. The tourists agreed to leave the bicycle at intermediate points between which one walks and the other rides. What will the mean speed of the tourists be? How long will the bicycle remain unused?

3. A man must get from point *A* on one bank of a river to point *B* on the other bank moving along straight line *AB* (Fig. 1). The width of the river *AC* = 1 km, the distance *BC* = 2 km, the maximum speed of the boat relative to the water *u* = 5 km/h and the river current velocity *v* = 2 km/h. Can the distance *AB* be covered in 30 minutes?

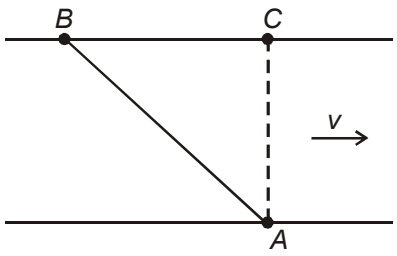



**Fig. 1**

4. A launch travels across a river from point *A* to point *B* on the opposite bank along a straight line *AB* forming an angle $\alpha$ with the bank (Fig. 2). The wind blows with a velocity of *u* at right angles to the bank. The flag on the mast of the launch forms an angle $\beta$ with the direction of its motion. Determine the speed of the launch with respect to the bank. Can the data in this problem be used to find the river current velocity?

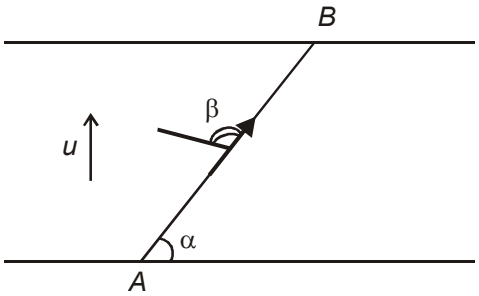



**Fig. 2**

5. Two intersecting straight lines move translationally in opposite directions with velocities $v_1$ and $v_2$ perpendicular to the corresponding lines. The angle between the lines is $\alpha$. Find the speed of the point of intersection of these lines.

6. Figures 3 and 4 show the velocity of a body and the change in its coordinate (parabola) with time. The origin of time reading coincides on both charts. Are the motions shown in the charts the same?

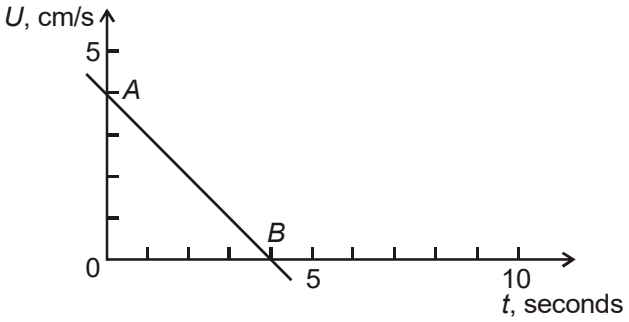



**Fig. 3**

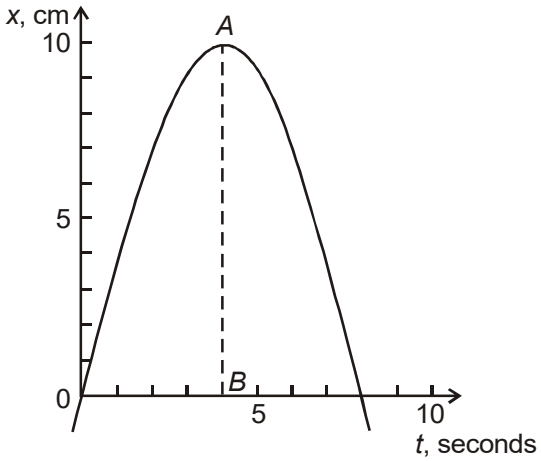



**Fig. 4**

7. Two steel balls freely drop onto an elastic plate, the first one from a height of $h_1$ = 44 cm and the second in $\tau > 0$ seconds after the first from a height of $h_2$ = 11 cm. After a certain time the velocities of the balls coincide in magnitude and direction. Find the time $\tau$ and the interval during which the velocities of the two balls remain the same. The balls do not collide.

8. An overhead travelling crane lifts a load from the ground with an upward acceleration of $a_1$. At the same time the hook of the crane carrying the load moves in a horizontal direction with an acceleration $a_2$ relative to the crane. Besides, the crane runs on its rails with a constant speed $V_0$ (Fig. 5). The initial speed of the hook relative to the crane is zero. Find the speed of the load with respect to the ground when it reaches the height $h$.

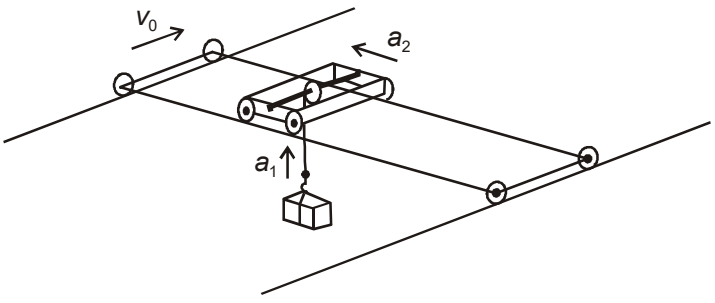



**Fig. 5**

9. Two pans with weights each equal to $G$ = 3 kgf are suspended from the ends of a string passed over two stationary pulleys. The string between the pulleys is cut and the free ends are connected to a dynamometer (Fig. 6). What will the dynamometer show? What weight $G_1$ should be added to one of the pans for the reading of the dynamometer not to change after a weight $G_2$ = 1 kgf is taken off the other pan? Disregard the masses of the pans, pulleys, thread and dynamometer.

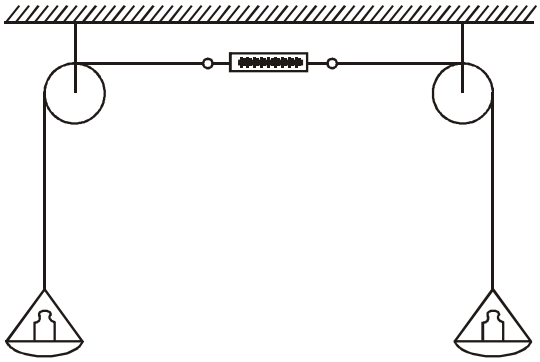


**Fig. 6**

10. A cart with a mass of $M$ = 500 g is connected by a string to a weight having a mass $m$ = 200 g. At the initial moment the cart moves to the left along a horizontal plane at a speed of $v_0$ = 7 m/s (Fig. 7). Find the magnitude and direction of the speed of the cart, the place it will be at and the distance it covers in $t$ = 5 seconds.

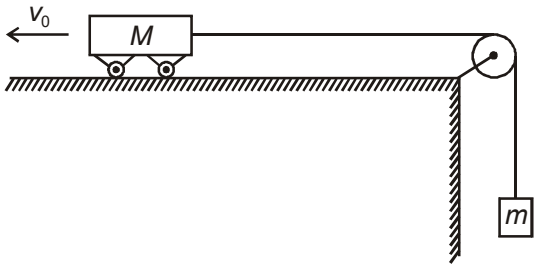



**Fig. 7**

11. A weightless string thrown over a stationary pulley is passed through a slit (Fig. 8). As the string moves it is acted upon by a constant friction force $F$ on the side of the slit. The ends of the string carry two weights with masses $m_1$ and $m_2$, respectively. Find the acceleration $a$ of the weights.

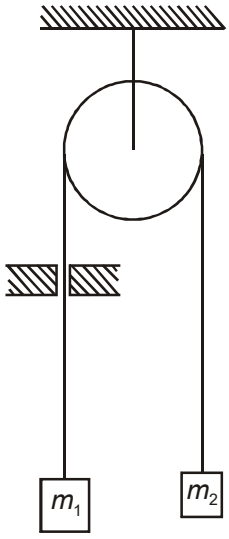



**Fig. 8**

12. A table with a weight of $G_1$ = 15 kgf can move without friction over a level floor. A weight of $G_2$ = 10 kgf is placed on the table, and a rope passed over two pulleys fastened to the table is attached to it (Fig. 9). The coefficient of friction between the weight and the table $k$ = 0.6. What acceleration will the table move with if a constant force of 8 kgf is applied to the free end of the rope? Consider two cases: (1) the force is directed horizontally, (2) the force is directed vertically upward.

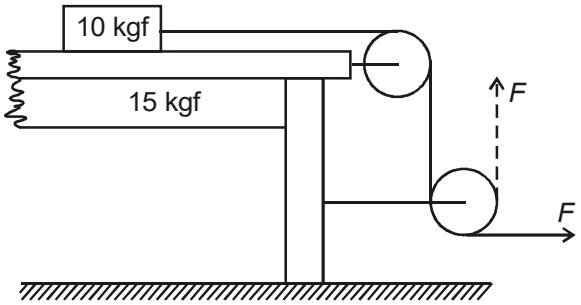



**Fig. 9**

13. A massive homogeneous cylinder that can revolve without friction around a horizontal axis is secured on a cart standing on a smooth level surface (Fig. 10). A bullet flying horizontally with a velocity $v$ strikes the cylinder and drops onto the cart. Does the speed of the cart that it acquires after the impact depend on the point where the bullet strikes the cylinder?

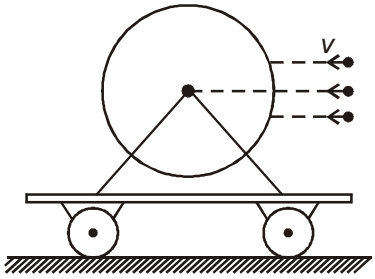



**Fig. 10**

14. A wedge with an angle $\alpha$ at the base can move without friction over a smooth level surface (Fig. 11). At what ratio between the masses $m_1$ and $m_2$ of the weights, that are connected by a string passed over a pulley, will the wedge remain stationary, and at what ratio will it begin to move to the right or left? The coefficient of friction between the weight of mass $m_2$ and the wedge is $k$.

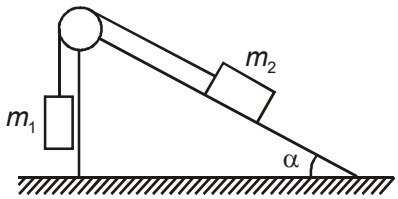



**Fig. 11**

15. Two identical weights are suspended from the ends of a string thrown over two pulleys (Fig. 12). Over what distance will a third weight of the same mass lower if it is attached to the middle of the string? The distance between the axes of the pulleys is $2\ell$. The friction in the axes of the pulleys is negligible.

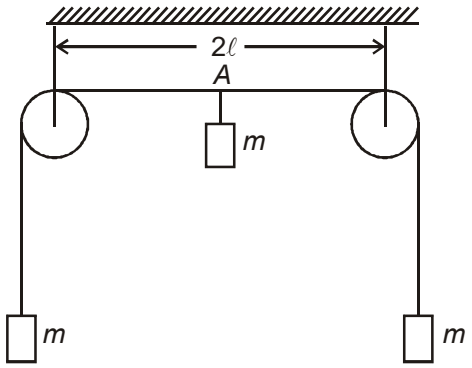



**Fig. 12**

16. A homogeneous board is balanced on a rough horizontal log (Fig. 14). After a weight has been added to one of the ends of the board, equilibrium can be obtained when the board forms an angle $\alpha$ with the horizon. What is the coefficient of friction between the board and the log?

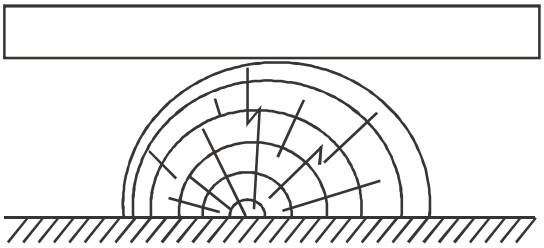


**Fig. 14**

17. A homogeneous thin rod *AB* with a length $\ell$ is placed onto the horizontal surface of a table. A string with a length of $2\ell$ is attached to end *B* of the rod (Fig. 15). How will the rod move if the other end *C* of the string is slowly lifted up a stationary vertical straight line *DO* passing through end *A* of the rod. Disregard the weight of the string.

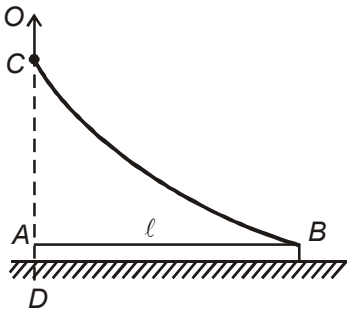



**Fig. 15**

18. A ladder with its centre of gravity at the middle stands on an absolutely smooth floor and leans against a smooth wall (Fig. 16). What should the tension of a rope tied to the middle of the ladder be to prevent its falling down?

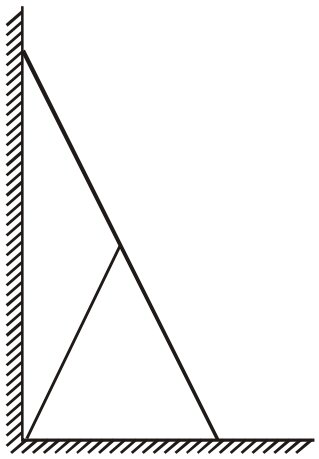


**Fig. 16**

19. A picture is attached to a vertical wall by means of string *AC* with a length $\ell$ forming an angle $\alpha$ with the wall. The height of the picture *BC* = *d* (Fig. 17). The bottom of the picture is not fastened. At what coefficient of friction between the picture and the wall will the picture be in equilibrium?

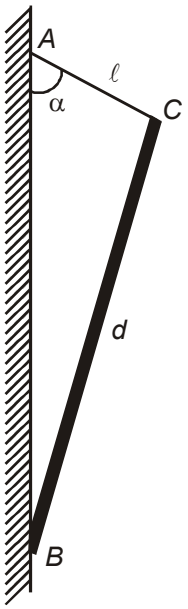



**Fig. 17**

20. A weightless unstretchable string is wound around a cylinder with a mass *m* (Fig. 18). With what minimum force $F_{min}$ and at what angle $\alpha_1$ to the horizon should the string be pulled for the rotating cylinder to remain in place? The coefficient of friction between the cylinder and the floor is *k*.

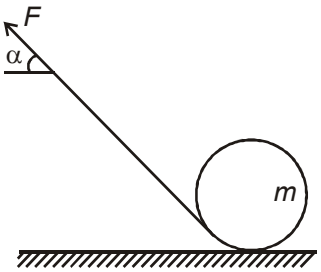



**Fig. 18**

21. Determine the centre of gravity of a thin homogeneous plate cut in the form of a sector with a radius *r* and a central angle $\alpha$ (Fig. 19).

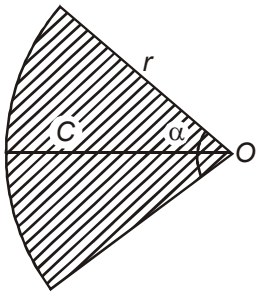



**Fig. 19**

22. Two elastic blocks of equal mass *m* and connected by a spring with a length $\ell$ rest on an absolutely smooth horizontal surface (Fig. 20). The coefficient of elasticity of the spring is *k*. A third block of the same mass *m* strikes the left-hand block with a velocity *v*. Prove that the blocks connected by the spring will always move in the same direction.

    Determine the velocities of the blocks when the spring is stretched as much as possible.

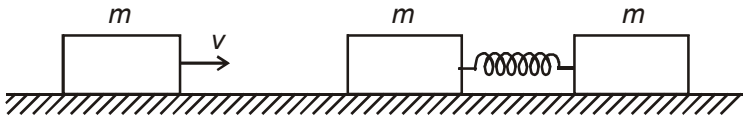



**Fig. 20**

23. A row of balls of identical size are placed at small intervals on a level surface (Fig. 21). The middle ball is made of steel and the others of ivory (the mass of the steel ball is greater). An ivory ball (of the same mass) strikes the balls from the right along the line of centres. How will the balls move after the impact?

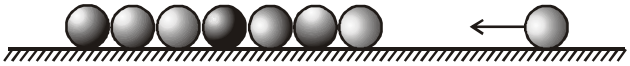


**Fig. 21**

24. Equal weights of mass *m* are suspended from the ends of a very long string thrown over two small stationary pulleys at a distance of $2\ell$ from each other (Fig. 22). Find the velocity of the weights after a sufficiently long time if a weight with a mass $2m$ is attached to the middle of the string.

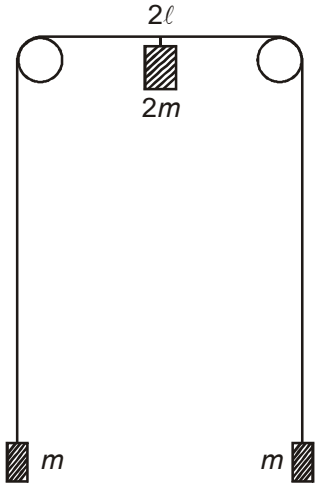



**Fig. 22**

25. A wagon with a mass *M* can move without friction along horizontal rails. A mathematical pendulum (a sphere with a mass *m* suspended from a long string $\ell$) is fastened on the wagon (Fig. 23). At the initial moment

the wagon and the pendulum were at rest and the string was deflected through an angle $\alpha$ from the vertical. What will the velocity of the wagon be when the pendulum string forms an angle $\beta$ ($\beta < \alpha$) with the vertical?

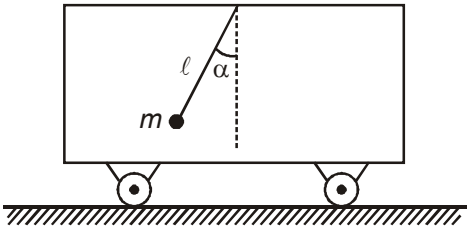



**Fig. 23**

26. Find the acceleration of body *A* which slides without initial velocity down a helical groove with a pitch *h* and a radius *R* at the end of the $n^{th}$ turn (Fig. 24). Disregard friction.

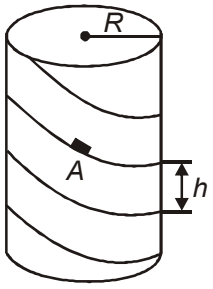



**Fig. 24**

27. A spool consists of a cylindrical core and two identical solid heads. The core rolls without slipping along a rough horizontal block with a constant velocity *v* (Fig. 25). The radius of the core is *r* and that of the heads is *R*.

What instantaneous velocity will points *A* and *B* on the rim of one of the heads have? What points on the heads have an instantaneous velocity equal in magnitude to the velocity of the spool core?

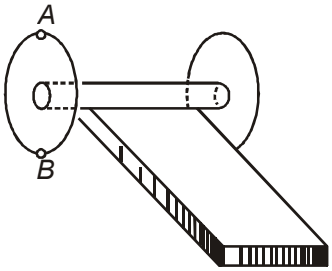



**Fig. 25**

28. A ball bearing supports the end of a shaft rotating with an angular velocity $\omega$. The diameter of the shaft axis is *d* and that of the race is *D* (Fig. 26). Find the linear velocity of motion of the centre of one of the balls if the race is stationary, or rotates with an angular velocity of $\Omega$. Assume that in both cases the balls do not slip when they roll over the shaft.

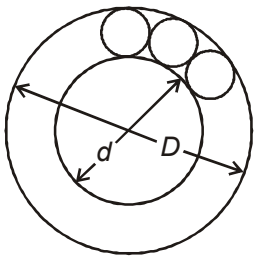



**Fig. 26**

29. Masses $m_1$ and $m_2$ are fastened to the ends of a weightless rod with a length $\ell$. The velocities of these

masses are in one plane and are equal to $v_1$ and $v_2$, respectively (Fig. 27). Find the velocity which the centre of gravity of the system moves with and the angular velocity which the rod rotates with relative to an axis passing through the centre of gravity.

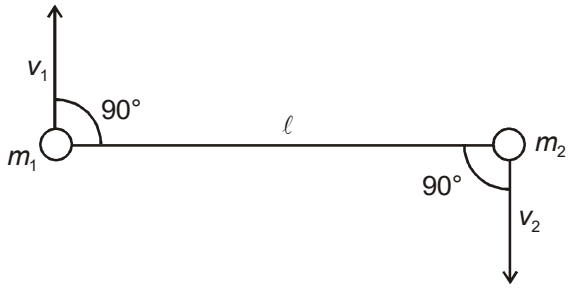



**Fig. 27**

30. A wire is bent along an arc with a radius $R$ (Fig. 28). A bead is placed on the wire that can move along it without friction. At the initial moment the bead was at point $O$. What horizontal velocity should be imparted to the bead for it to get onto the wire again at point $B$ after fiying a certain distance ($AB$) through the air?

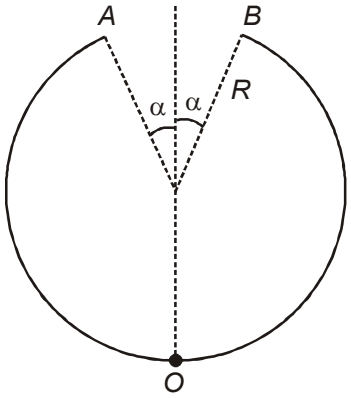



**Fig. 28**

31. A small body slides down an inclined surface passing into a loop from the minimum height ensuring that the body does not leave the surface of the loop (Fig. 29). What symmetrical segment with an angle $\alpha < 90°$ can be cut out of the loop for the body to reach point $B$ after travelling a certain distance in the air?

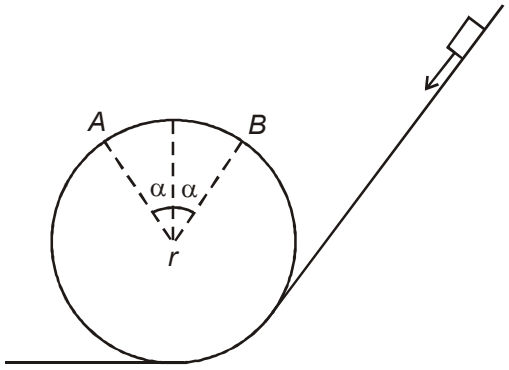



**Fig. 29**

How will the body move if the angle $\alpha$ is greater or smaller than the found one? Disregard friction and the resistance of the air.

32. A weightless rod can rotate in a vertical plane with respect to point *O*. The rod carries masses $m_1$ and $m_2$ at distances $r_1$ and $r_2$ from *O* (Fig. 30). The rod is released without any initial velocity from a position forming an angle $\alpha$ with the vertical. Determine the linear velocities of masses $m_1$ and $m_2$ when the rod reaches a vertical position.

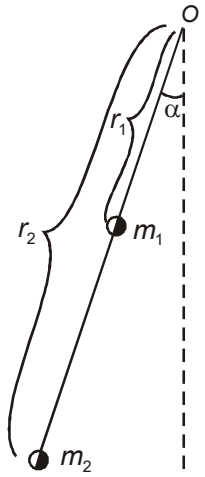



**Fig. 30**

33. A rigid rod bent as shown in Fig. 31 rotates with an angular velocity $\omega$ around axis *OO′*. A weight with a mass *m* is attached to the end of the rod. Find the force which the rod acts with on the mass *m*.

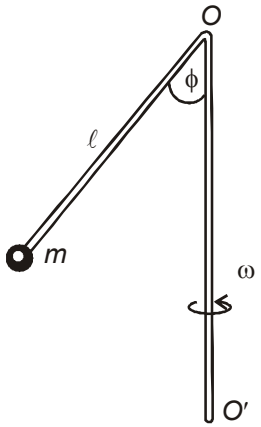



**Fig. 31**

34. A weightless rod carries two weights of mass *m* and *M*. The rod is hinge-jointed to vertical axis *OO′* (Fig. 32), which rotates with an angular velocity $\omega$. Determine the angle $\phi$ formed by the rod and the vertical.

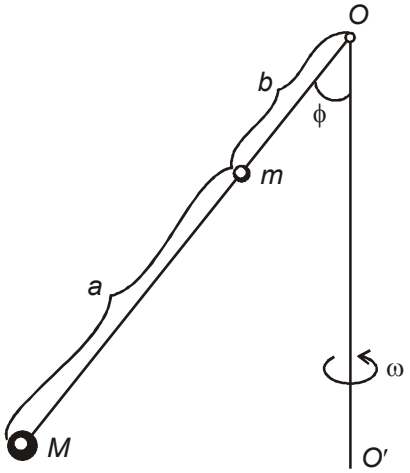



**Fig. 32**

35. A horizontal straight bar rotates with a constant angular velocity around a vertical axis. A body can move without friction over the bar. Initially, the body is retained in equilibrium by a spring (Fig. 33). What will happen to the body if an initial velocity is imparted to it along the bar? The length of the spring in an unstretehed state can be neglected.

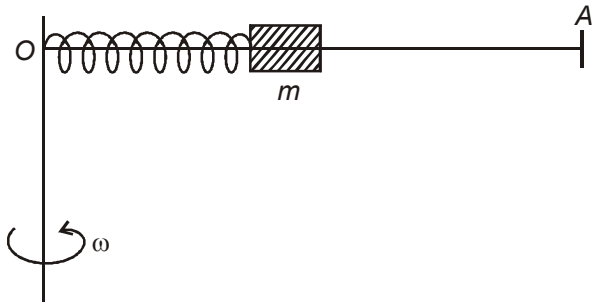



**Fig. 33**

36. A heavy spool with a thread wound on it lies on a rough horizontal surface over which it can roll without slipping. If the thread is pulled to the left in a horizontal direction, the spool will also move to the left. If the direction of the thread is changed (Fig. 34), the spool will begin to roll to the right at a certain angle $\alpha$ between the direction of the thread and a vertical line.

Determine this angle. What will happen to the spool at the given value of this angle? The radius of the spool heads is *R* and that of its core is *r*.

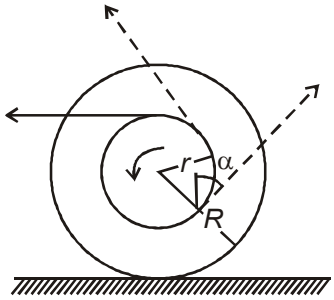



**Fig. 34**

37. A cylindrical tube with a radius *r* is connected by means of spokes to two hoops with a radius *R*. The mass of both the hoops is *M*. The mass of the tube and the spokes in comparison with the mass *M* can be neglected. A string passed over a weightless pulley is wound around the tube. A weight with a mass *m* is attached to the end of the string (Fig. 35).

Find the acceleration of the weight, the tension of the string and the force of friction acting between the hoops and the surface. (Assume that the hoops do not slip.) Also determine the coefficient of friction at which the hoops will slip.

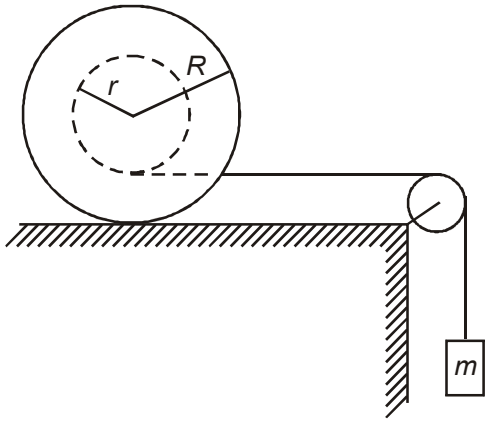



**Fig. 35**

38. A lead ball with a radius of $R$ = 50 cm has inside a spherical space with a radius $r$ = 5 cm whose centre is at a distance of $d$ = 40 cm from the centre of the ball (Fig. 36). With what force will a material particle with a mass of $m$ = 10 g at a distance of $\ell$ = 80 cm from the centre of the ball be attracted to it if the line connecting the centres of the ball and the space forms an angle $\alpha$ = 60° with the line connecting the centre of the ball and the material particle?

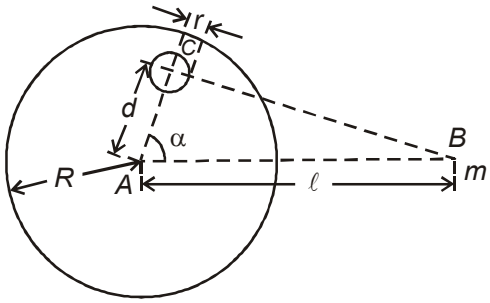



**Fig. 36**

39. A light pendulum consisting of a rod and a disk (Fig. 37) is attached to a wooden block which can freely fall along guiding wires.

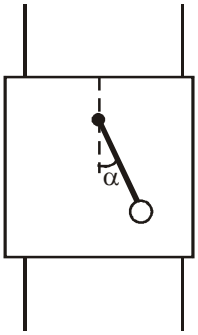



**Fig. 37**

The pendulum is brought out of equilibrium through an angle $\alpha$ and released. At the moment when the pendulum passed through its lowermost position, the block was released and began to fall freely. How will the pendulum move with respect to the block? Disregard friction and the resistance of the air.

40. Two air bubbles with a radius $r$ are present in water. Are the bubbles attracted or repulsed? What is the force of interaction between them? The distance between the bubbles is $R$.

41. A tube floats vertically in water (Fig. 38). The portion of the tube protruding from the water is $h$ = 5 cm. Oil with a specific weight of $\gamma$ = 0.9 gf/cm$^3$ is poured into the tube. What length can the tube have so that it can be completely filled with oil?

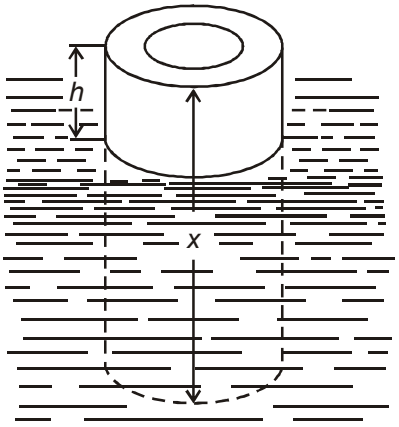



**Fig. 38**

42. A cart supports a cubic tank filled with water up to its top (Fig. 39). The cart moves with a constant acceleration *a*. Determine the pressure at point *A* which is at a depth *h* and a distance $\ell$ from the front wall, if the tank is tightly closed with a lid. In uniform motion the lid does not exert any pressure on the water.

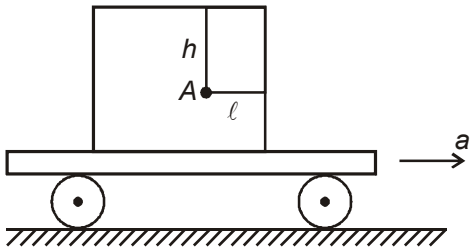



**Fig. 39**

## 2. HEAT MOLECULAR PHYSICS

1. A clock with a metallic pendulum is $\tau_1$ = 5 seconds fast each day at a temperature of $t_1$ = +15°C and $\tau_2$ = 10 seconds slow at a temperature of $t_2$ = + 30°C. Find the coefficient $\alpha$ of thermal expansion of the pendulum metal, remembering that the period of oscillations of a pendulum $T = 2\pi\sqrt{\frac{\ell}{g}}$, where $\ell$ is the length of the pendulum and *g* is the gravity acceleration.

2. It is known that if water is heated or cooled with certain care it can retain its liquid state at temperatures below 0°C and higher than +100°C.

   A calorimeter with a heat capacity of *q* = 400 cal/deg contains $m_1$ = 1 kg of water cooled to $t_1$ = –10°C. Next $m_2$ = 100 g of water overheated to $t_2$= +120°C is added to it.

   What is the temperature in the calorimeter?

3. A wall consists of two adjoining panels made of different materials. The coefficients of thermal conductivity and the thicknesses of the panels are $\lambda_1$, $d_1$ and $\lambda_2$, $d_2$, respectively (Fig. 1). The temperatures of the external surfaces of the wall are $T_1$ and $T_0$ (where $T_0 > T_1$) and are kept constant. Find the temperature $T_2$ on the plane between the panels.

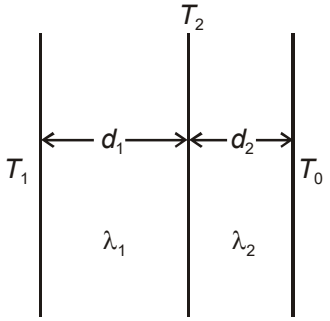



**Fig. 1**

4. A wall consists of alternating blocks with a length *d* and coefficients of thermal conductivity $\lambda_1$ and $\lambda_2$ (Fig. 2). The cross-sectional areas of the blocks are the same. Determine the coefficient of thermal conductivity of the wall.

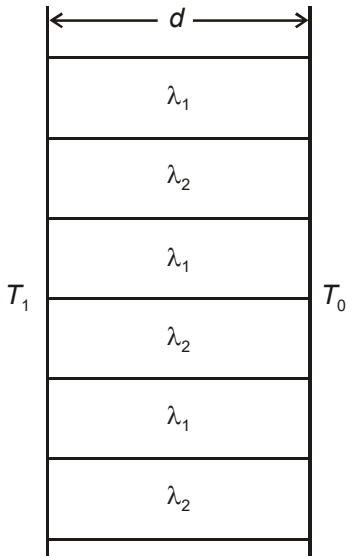



**Fig. 2**

5. Twenty grammes of helium in a cylinder under a piston are transferred infinitely slowly from a state with a volume of $V_1$ = 32 liter and a pressure of $P_1$ = 4.1 atm to a state with $V_2$ = 9 liter and $P_2$ = 15.5 atm. What maximum temperature will the gas reach in this process if it is depicted on the *P*-*V* diagram as a straight line (Fig. 3)?

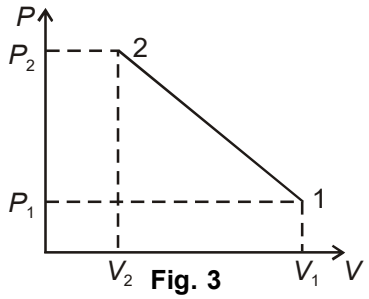



**Fig. 3**

6. A capillary tube is lowered into a vessel with a liquid whose vapour pressure may be neglected. The density of the liquid is $\rho$. The vessel and the tube are in a vacuum under the bell of an air pump (Fig. 4). Find the pressure inside the liquid in the capillary tube at a height *h* above the level of the liquid in the vessel.

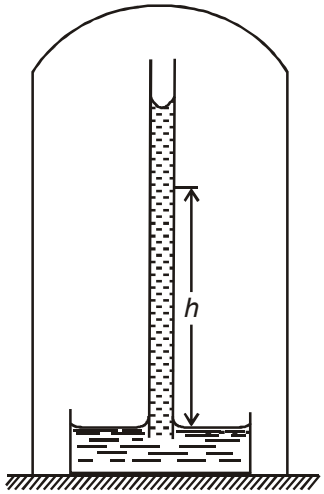



**Fig. 4**

7. A calorimeter contains 400 g of water at a temperature of +5°C. Then, 200 g of water at a temperature of +10°C are added and 400 g of ice at a temperature of –60°C are put in. What is the temperature in the calorimeter?

8. A wire with a length of $2\ell$ is stretched between two posts. A lantern with a mass *M* is suspended exactly from the middle of the wire. The cross-sectional area of the wire is *A* and its modulus of elasticity *E*. Determine the angle $\alpha$ of sagging of the wire, considering it to be small (Fig. 5).

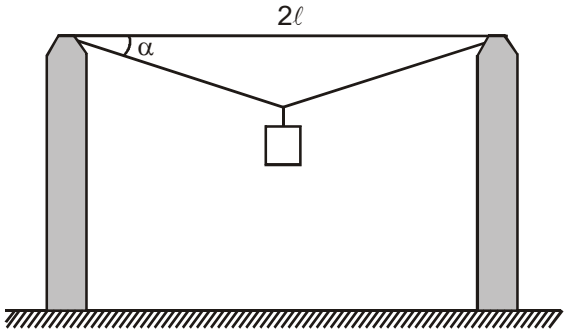



**Fig. 5**

9. A homogeneous block with a mass *m* = 100 kg hangs on three vertical wires of equal length arranged symmetrically (Fig. 6). Find the tension of the wires if the middle wire is of steel and the other two are of copper. All the wires have the same cross section. Consider the modulus of elasticity of steel to be double that of copper.

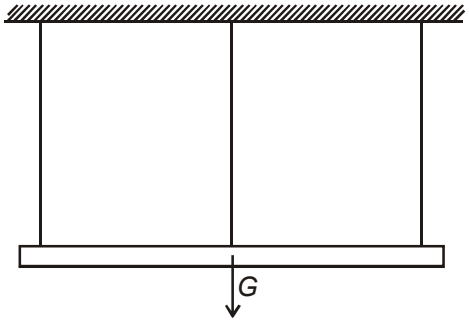



**Fig. 6**

## 3. ELECTRICITY AND MAGNETISM

1. Each edge of a cube made of wire contains a capacitor *C* (Fig. 3). Find the capacitance of this battery if it is connected to the circuit by means of conductors joined to the opposite apices *A* and *B* of the cube.

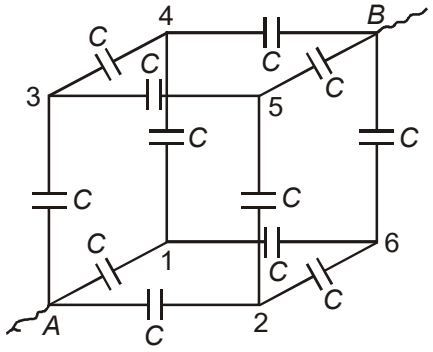



**Fig. 3**

2. A dielectric in the form of a sphere is introduced into a homogeneous electric field. How will the intensity of the field change at points *A*, *B* and *C* (Fig. 4).

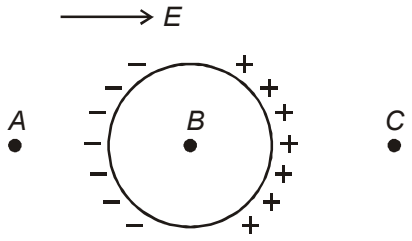



**Fig. 4**

3. The space between the plates of a plane capacitor is filled with a dielectric as shown in Fig. 5. The area of each plate is $A$ and the permittivity of the dielectric is $\varepsilon_r$. Find the capacitance of the capacitor in both cases.

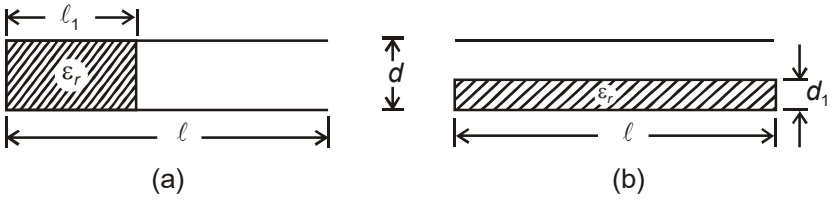



**Fig. 5**

4. Wires identical in cross section $A$ and resistivity $\rho$ pare soldered into a rectangle $ADBC$ (Fig. 6) with the diagonal $AB$ of the same cross section and material. Find the resistance between points $A$ and $B$ and between $C$ and $D$ if $AD = BC = a$ and $AC = BD = b$.

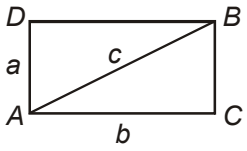



**Fig. 6**

5. A charge $Q$ is imparted to two identical plane capacitors connected in parallel. At the moment of time $t = 0$ the distance between the plates of the first capacitor begins to increase uniformly according to the law $d_1 = d_0 + vt$, and the distance between the plates of the second capacitor to decrease uniformly according to the law $d_2 = d_0 - vt$. Neglecting the resistance of the feeding wires. find the intensity of the current in the circuit when the plates of the capacitors move.

6. A storage battery with an e.m.f. of $\varepsilon = 24$ V was connected to one end of a two-conductor telephone line with a length of $L = 5.6$ km to find an insulation breakdown between the wires. It was found that if the conductors at the other end of the line were opened, the current flowing through the battery was $I_1 = 1.5$ A, and if short-circuited the current was $I_2 = 2$ A. The short-circuit current of the battery $I_3 = 96$ A and the resistance of each conductor of the line $r = 7\ \Omega$. Find the resistance of the insulation $R$ at the point of breakdown.

7. An electric lamp with a tungsten filament consumes 50 watts. When the lamp burns, the temperature of its filament is 2,500°C. What power will be consumed by the lamp at the first moment after it is switched on? The temperature coefficient of resistance of tungsten $\alpha = 4.5 \times 10^{-3}$ deg$^{-1}$

8. A charge $+Q$ is uniformly distributed over a thin ring with a radius $R$. Find the velocity of a negative point charge $-Q$ at the moment it passes through the centre $O$ of the ring if the charge $-Q$ was initially at rest at point $A$ sufficiently removed from the ring (Fig. 9). The mass of the charge $-Q$ is equal to $m$. The ring is stationary.

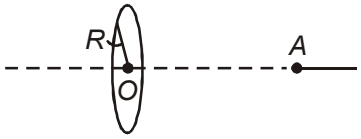



**Fig. 9**

9. A current flows through a conductor arranged in one plane as shown in Fig. 10. Find the intensity of the magnetic field at an arbitrary point on line *AB*, which is the axis of symmetry of the conductor.

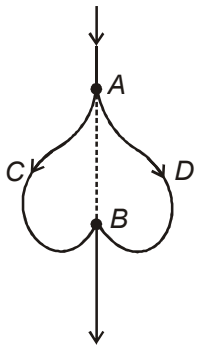



**Fig. 10**

10. Remembering that the intensity of a magnetic field inside a long cylindrical conductor $H = k2\pi jr$, where $j$ is the current density and $r$ is the distance from the conductor axis, find the intensity of the field at an arbitrary point on a long cylindrical space inside the conductor (Fig. 11) through which a current with a density $j$ flows. The axis of, the space is parallel to the axis of the conductor and is at a distance $d$ from it.

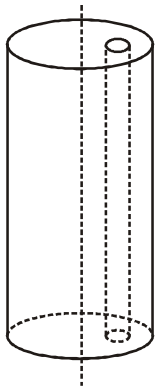


**Fig. 11**

11. A rectangular circuit *ABCD* with sides *a* and *b* placed into a homogeneous magnetic field with an intensity *H* can revolve around axis *OO′* (Fig. 12). A direct current *I* constantly flows through the circuit.

    Determine the work performed by the magnetic field when the circuit is turned through 180° if initially the plane of the circuit is perpendicular to the magnetic field and arranged as shown in Fig. 12.

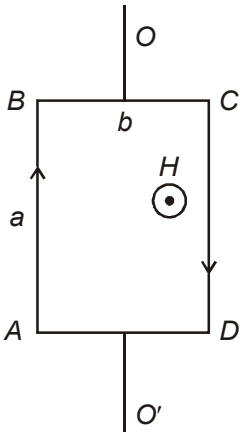



**Fig. 12**

12. A current *I* flows along a metal band with a width *AB* = *a* placed in a magnetic field with an intensity *H* perpendicular to the band (Fig. 13). Find the potential difference between points *A* and *B* of the band.

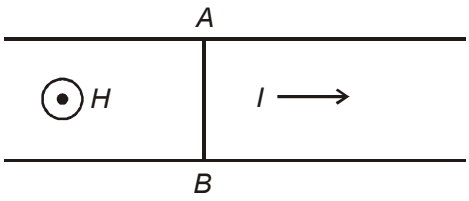



**Fig. 13**

13. An uncharged metal block has the form of a rectangular parallelepiped with sides *a*, *b*, and *c*(*a* >> *c*, *b* >> *c*). The block moves in a magnetic field in the direction of side *a* with a velocity *v*. The intensity of the magnetic field *H* is perpendicular to the base of the block with the sides *a* and *c* (Fig. 14).

    Determine the intensity of the electric field in the block and the density of the electric charges on the surfaces of the parallelepiped formed by sides *a* and *b*.

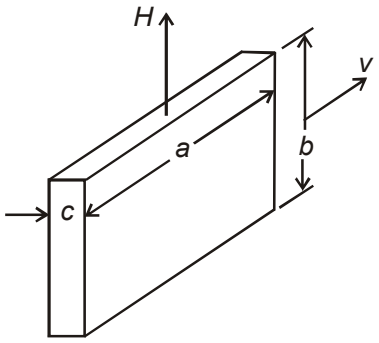



**Fig. 14**

14. A rectangular circuit *ABCD* moves translationally in the magnetic field of a current flowing along straight long conductor *OO′* (Fig. 15). Find the direction of the current induced in the circuit if the turn moves away from the conductor.

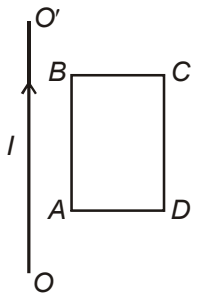



**Fig. 15**

15. A ring of a rectangular cross section (Fig. 16) is made of a material whose resistivity is $\rho$, The ring is placed in a homogeneous magnetic field. The intensity of the magnetic field is directed along the axis of the ring and increases directly with time, $H = kt$. Find the intensity of the current induced in the ring.

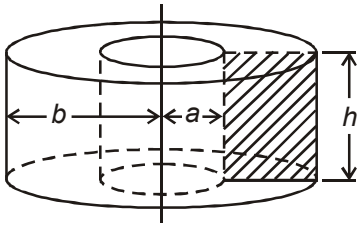



**Fig. 16**

16. Determine the current in the conductors of the circuit shown in Fig. 17 if the intensity of a homogeneous magnetic field is perpendicular to the plane of the drawing and changes in time according to the law $H = kt$. The resistance of a unit of length of the conductors is $r$.

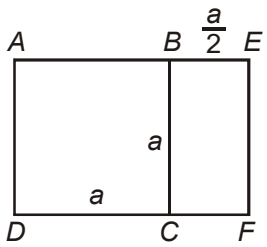



**Fig. 17**

17. A variable magnetic field creates a constant e.m.f. $\varepsilon$ in a circular conductor *ADBKA*. The resistances of the conductors *ADB*, *AKB* and *ACB* (Fig. 18) are equal to $R_1$, $R_2$ and $R_3$, respectively. What current will be shown by ammeter *C*? The magnetic field is concentrated near the axis of the circular conductor.

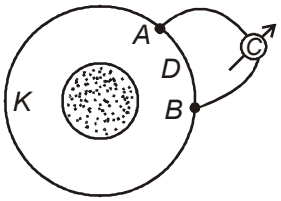



**Fig. 18**

18. The resistance of conductor *ACB* (see problem 22) is $R_3 = 0$. Find the currents $I_1$, $I_2$, and $I_3$ and the potential difference $U_A – U_B$.

19. A conductor with a length $\ell$ and mass $m$ can slide without friction along two vertical racks *AB* and *CD* connected by a resistor *R*. The system is in a homogeneous magnetic field whose intensity *H* is perpendicular

to the plane of the drawing (Fig. 19).

How will the movable conductor travel in the field of gravity if the resistance of the conductor itself and the racks is neglected?

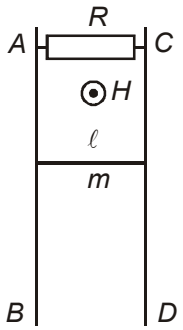



**Fig. 19**

20. A conductor with a mass *m* and length $\ell$ can move without friction along two metallic parallel racks in a horizontal plane and connected across capacitor *C*. The entire system is in a homogeneous magnetic field whose intensity *H* is directed upward. A force *F* is applied to the middle of the conductor perpendicular to it and parallel to the racks (Fig. 20).

Determine the acceleration of the conductor if the resistance of the racks, feeding wires and conductor is zero. What kinds of energy will the work of the force *F* be converted into? Assume that the velocity of the conductor is zero at the initial moment.

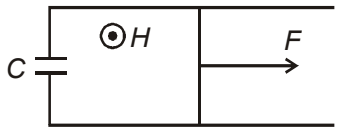



**Fig. 20**

21. A wire ring secured on the axis passing through its centre and perpendicular to the force lines is placed in a homogeneous magnetic field (Fig. 21). The intensity of the field begins to grow. Find the possible positions of equilibrium of the ring and show the position of stable equilibrium. What will happen if the intensity of the field decreases?

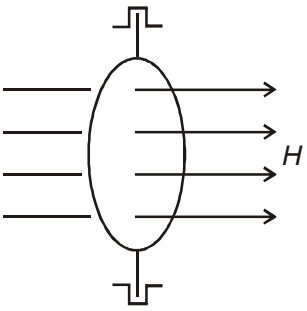



**Fig. 21**

22. A ring made of a superconductor is placed into a homogeneous magnetic field whose intensity grows from zero to $H_0$. The plane of the ring is perpendicular to the force lines of the field. Find the intensity of the induction current appearing in the ring. The radius of the ring is *r*, and its inductance *L*.

# 4. OSCILLATIONS AND WAVES

1. A hoop with a mass $m$ and a radius $r$ can roll without slipping over the internal surface of a cylinder with a radius $R$ (Fig. 2). Determine the period of motion of the hoop centre if the angle $\phi$ is small.

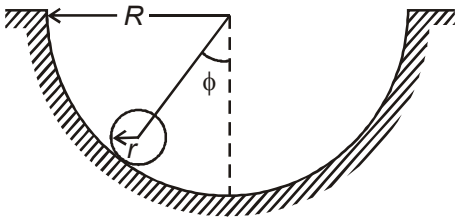



**Fig. 2**

2. Find the period of oscillations of the pendulum shown in Fig. 3. Consider the rod carrying the masses $m_1$ and $m_2$ to be weightless.

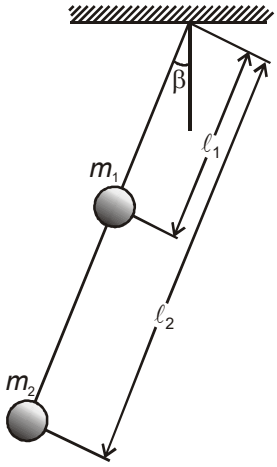



**Fig. 3**

3. Find the period of oscillations of a pendulum made of a thin homogeneous half-ring with a radius $r$ suspended on weightless strings $OA$ and $OB$ as shown in Fig. 4.

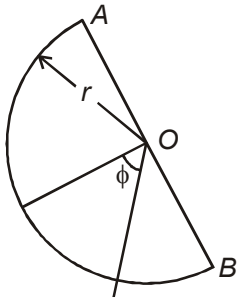



**Fig. 4**

4. A string secured at its ends is stretched with a force $f$. A point weight with a mass $m$ is attached to the middle of the string (Fig. 5). Determine the period of small oscillations of the weight. The mass of the string and the force of gravity can be neglected.

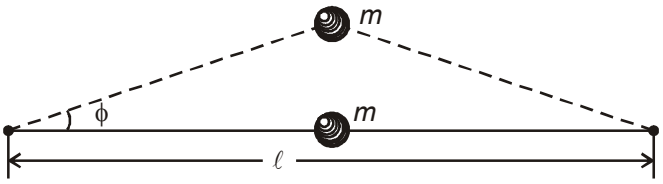



**Fig. 5**

5. Two mathematical pendulums each with a length $\ell$ are connected by a weightless spring as shown in Fig. 6. The coefficient of elasticity of the spring is *k*. In equilibrium the pendulums occupy a vertical position and the spring is not deformed. Determine the frequencies of small oscillations of the two linked pendulums when they are deflected in one plane through equal angles in one direction (oscillations in phase) and in opposite directions (oscillations in antiphase).

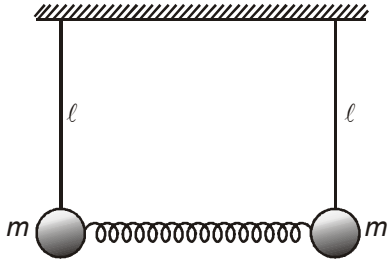



**Fig. 6**

## 5. GEOMETRICAL OPTICS

1. A flat mirror revolves at a constant angular velocity, making $n = 0.5$ revolution per second. With what velocity will a light spot move along a spherical screen with a radius of 10 metres if the mirror is at the centre of curvature of the screen?
2. A narrow beam of light *S* is incident on a dihedral angle $\alpha = 60°$ formed by identical flat mirrors *OM* and *ON* secured on axis *O* (Fig. 2). After being reflected from the mirrors, the light is focussed by lens *L* and gets into stationary receiver *R*. The mirrors rotate with a constant angular velocity.

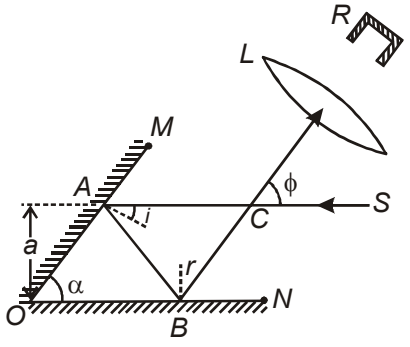



**Fig. 2**

What part of the light energy of the beam will reach the receiver during a time that greatly exceeds the period of rotation if the beam passes at *a* distance a from an axis equal to half the length of mirror *OM*?

3. A cubical vessel with non-transparent walls is so located that the eye of an observer does not see its bottom, but sees all of the wall *CD* (Fig. 4).

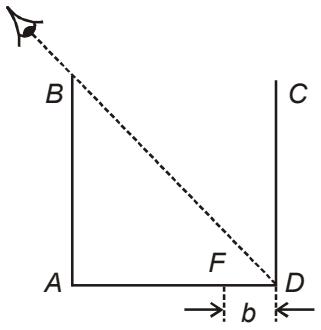



**Fig. 4**

What amount of water should be poured into the vessel for the observer to see an object $F$ arranged at a distance of $b$ = 10 cm from corner $D$? The face of the vessel is $a$ = 40 cm.

4. A convexo-convex lens made of glass with a refraction index of $n$ = 1.6 has a focal length of $f$ = 10 cm. What will the focal length of this lens be if it is placed into a transparent medium with a refraction index of $n_1$ = 1.5? Also find the focal length of this lens in a medium with a refraction index of $n_2$ = 1.7.

5. A thin glass lens has an optical power of $D$ = 5 diopters. When this lens is immersed into a liquid with a refraction index $n_2$ the lens acts as a divergent one with a focal length of $f$ = 100 cm. Find the refraction index $n_2$ of the liquid if that of the lens glass is $n_1$ = 1.5.

6. Find by construction the optical centre of a lens and its main foci on the given optical axis $N_1N_2$ if the positions of the source $S$ and the image $S'$ are known (Fig. 5).




**Fig. 5**

7. An object $AB$ is at a distance of $a$ = 36 cm from a lens with a focal length of $f$ = 30 cm. A flat mirror turned through 45° with respect to the optical axis of the lens is placed behind it at a distance of $\ell$ = 1 metre (Fig. 6).

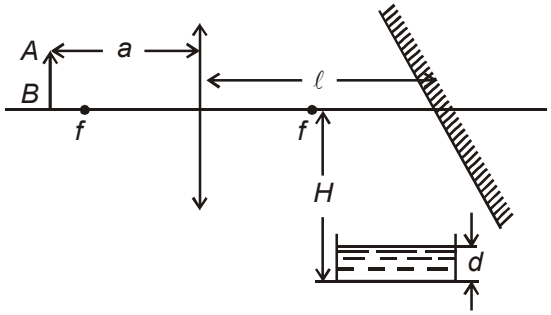



**Fig. 6**

At what distance $H$ from the optical axis should the bottom of a tray with water be placed to obtain a sharp image of the object on the bottom? The thickness of the water layer in the tray is $d$ = 20 cm.

8. A parallel beam of light is incident on a system consisting of three thin lenses with a common optical axis. The focal lengths of the lenses are equal to $f_1$ = + 10 cm, $f_2$ = –20 cm and $f_3$ = +9 cm, respectively. The distance between the first and the second lenses is 15 cm and between the second and the third 5 cm. Find the position of the point at which the beam converges when it leaves the system of lenses.

9. A short-sighted man, the accommodation of whose eye is between $a_1$ = 12 cm and $a_2$ = 60 cm, wears spectacles through which he distinctly sees remote objects. Determine the minimum distance $a_3$ at which the man can read a book through his spectacles.

## 6. PHYSICAL OPTICS

1. Two flat mirrors form an angle close to 180° (Fig. 1). A source of light *S* is placed at equal distances *b* from the mirrors. Find the interval between adjacent interference bands on screen *MN* at a distance *OA* = *a* from the point of intersection of the mirrors. The length of the light wave is known and equal to $\lambda$. Shield *C* does not allow the light to pass directly from the source to the screen.

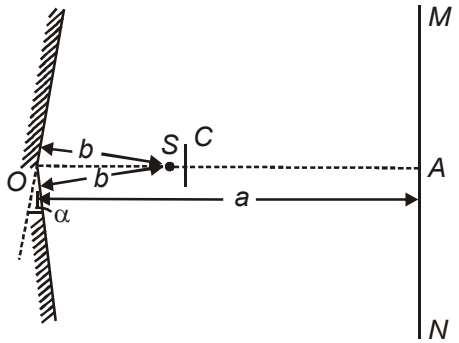



**Fig. 1**

2. Lloyd's interference experiment consisted in obtaining on a screen a pattern from source *S* and from its virtual image *S′* in mirror *AO* (Fig. 2). How will the interference pattern obtained from sources *S* and *S′* differ from the pattern considered in by two coherent sources?

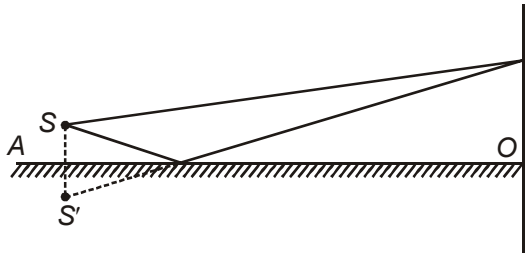



**Fig. 2**

3. Two point sources with the same phases of oscillation are on a straight line perpendicular to a screen. The nearest source is at a distance of $D >> \lambda$ from the screen. What shape will the interference bands have on the screen? What is the distance on the screen from the perpendicular to the nearest bright band if the distance between the sources is $\ell = n\lambda >> \lambda$ ($n$ is an integer)?

4. Light from source *S* is incident on the Fresnel biprism shown in Fig. 3. The light beams refracted by the different faces of the prism partly overlap and produce an interference pattern on a screen on its section *AB*.

   Find the distance between adjacent interference bands if the distance from the source to the prism is *a* = 1 metre and from the prism to the screen *b* = 4 metres. The angle of refraction of the prism is $\alpha = 2 \times 10^{-3}$ rad.

   The glass which the prism is made of has a refraction index of *n* = 1.5. The length of the light wave $\lambda$ = 6,000 Å.

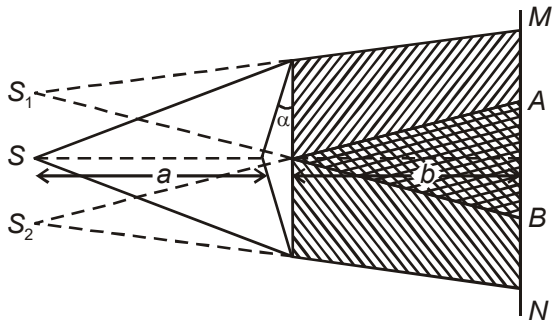



**Fig. 3**

5. The following method is used to facilitate the fabrication of a biprism with an angle close to 180° (see Problem 4). A biprism with an angle $\beta$ that appreciably differs from 180° is placed into a vessel filled with a liquid having a refraction index $n_1$, or serves as one of the walls of this vessel (Fig. 4).

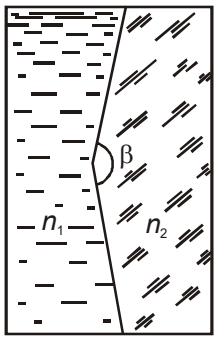



**Fig. 4**

Calculate the angle $\delta$ of an equivalent biprism in air. The refraction index of the prism substance. is $n_2$.

Perform the calculations for $n_1$ = 1.5 (benzene), $n_2$ = 1.52 (glass), $\beta$ = 170°.

6. Why are Newton's rings formed only by the interference of rays 2 and 3 reflected from the boundaries of the air layer between the lens and the glass (Fig. 5), while ray 4 reflected from the flat face of the lens does not affect the nature of the interference pattern?

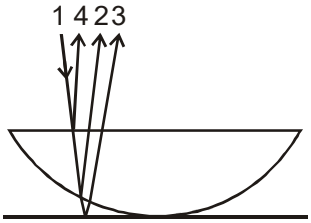


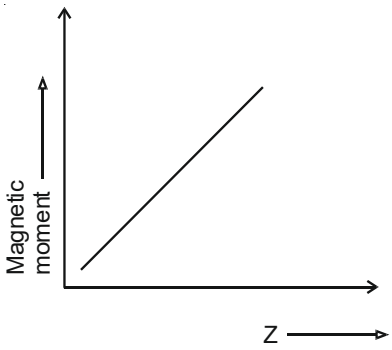

**Fig. 5**

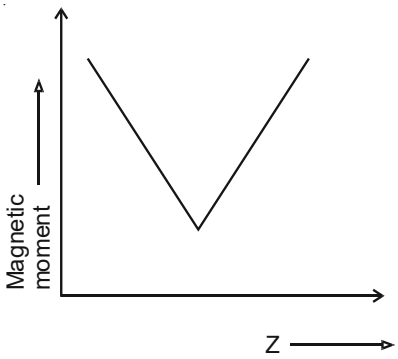
7. A plane-convex lens with a radius of curvature $R_2 < R_1$ is placed with its convex side onto the surface of a plane-concave lens having a radius of curvature of $R_1$. Find the radii of Newton's rings that appear around the point of contact of the lenses if monochromatic light with a wavelength $\lambda$, is normally incident onto the system.

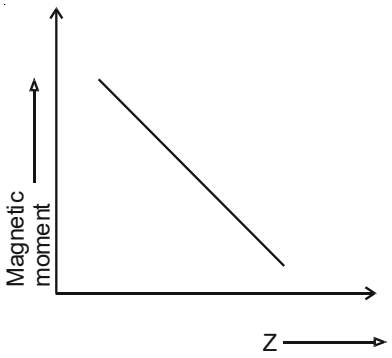
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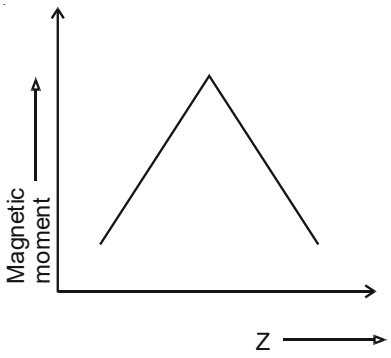
# Chemistry

1. Which of the following graph is correct representation between atomic number (Z) and magnetic moment of *d*-block elements? [Outer electronic configuration : $(n-1)d^x ns^{1 \text{ or } 2}$]

(A) 


(B) 


(C) 


(D) 


2. Which of the following transformation least energy is required?

(A) $F^-_{(g)} \rightarrow F_{(g)} + e^-$

(B) $P^-_{(g)} \rightarrow P_{(g)} + e^-$

(C) $S^-_{(g)} \rightarrow S_{(g)} + e^-$

(D) $Cl^-_{(g)} \rightarrow Cl_{(g)} + e^-$

3. Which of the following order is incorrect?

(A) Electronegativity of central atom : $CF_4 > CH_4 > SiH_4$

(B) Hydration energy : $Al^{3+} > Be^{2+} > Mg^{2+} > Na^+$

(C) Electrical conductance : $F^-_{(aq.)} > Cl^-_{(aq.)} > S^{2-}_{(aq.)}$

(D) Magnetic moment : $Ni^{4+} > V^{3+} > Sr^{2+}$

4. The ground state electronic configurations of the elements, U, V, W, X and Y (these symbols do not have any chemical significance) are as follows:

U $1s^2 2s^2 2p^3$

V $1s^2 2s^2 2p^6 3s^1$

W $1s^2 2s^2 2p^6 3s^2 3p^2$

X $1s^2 2s^2 2p^6 3s^2 3p^6 3d^5 4s^2$

Y $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^6$

Determine which sequence of elements satisfy the following statement :

(i) Element forms a carbonate which is not decomposed by heating

(ii) Element is most likely to form coloured ionic compounds

(iii) Element has larger atomic radius

(iv) Element forms only acidic oxide

(A) V W Y U (B) V X Y W

(C) V W Y X (D) V X W U

5. A, B and C are hydroxy-compounds of the elements X, Y and Z respectively. X, Y and Z are in the same period of the periodic table. A gives an aqueous solution of pH less than seven. B reacts with both strong acids and strong alkalis. C gives an aqueous solution which is strongly alkaline.

Which of the following statements is/are true?

(i) The three elements are metals.

(ii) The electronegativities decrease from X to Y to Z.

(iii) The atomic radius decreases in the order X, Y and Z.

(iv) X, Y and Z could be phosphorus, aluminium and sodium respectively.

(A) Only i, ii, iii correct (B) Only i, iii correct

(C) Only ii, iv correct (D) Only ii, iii, iv correct

6. Consider the following ionisation reactions:

I. E. (kJ $mol^{-1}$) I.E. (kJ $mol^{-1}$)

$A_{(g)} \longrightarrow A^+_{(g)} + e^-$, $A_1$ $B_{(g)} \longrightarrow B^+_{(g)} + e^-$, $B_1$

$B^+_{(g)} \longrightarrow B^{2+}_{(g)} + e^-$, $B_2$ $C_{(g)} \longrightarrow C^+_{(g)} + e^-$, $C_1$

$C^+_{(g)} \longrightarrow C^{2+}_{(g)} + e^-$, $C_2$ $C^{2+}_{(g)} \longrightarrow C^{3+}_{(g)} + e^-, C_3$

If monovalent positive ion of A, divalent positive ion of B and trivalent positive ion of C have zero electron. Then incorrect order of corresponding I.E. is:

(A) $C_3 > B_2 > A_1$

(B) $B_1 > A_1 > C_1$

(C) $C_3 > C_2 > B_2$

(D) $B_2 > C_3 > A_1$

7. First three ionisation energies (in kJ/mol) of three representative elements are given below:

| Element | $IE_1$ | $IE_2$ | $IE_3$ |
|---|---|---|---|
| P | 495.8 | 4562 | 6910 |
| Q | 737.7 | 1451 | 7733 |
| R | 577.5 | 1817 | 2745 |

Then incorrect option is :

(A) Q : Alkaline earth metal

(B) P : Alkali metals

(C) R : s-block element

(D) They belong to same period

8. Aqueous solutions of two compounds $M_1 - O - H$ and $M_2 - O - H$ are prepared in two different beakers. If, the electronegativity of $M_1$ = 3.4, $M_2$ = 1.2, O = 3.5 and H = 2.1, then the nature of two solutions will be respectively:

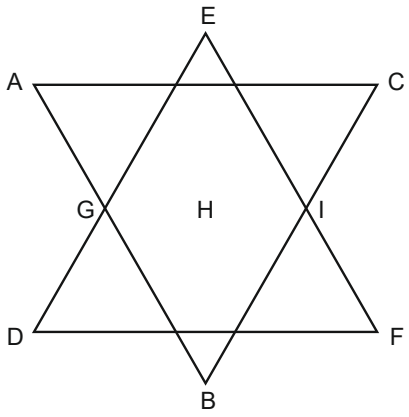
(A) Acidic, basic

(B) Acidic, acidic

(C) Basic, acidic

(D) Basic, basic

**Paragraph for Q. Nos. 9 to 11**




There are nine elements A to I. These are belongs to p-block element other than halogen. If atomic number of B is average of atomic number of A and C and atomic number of E is average of atomic number of D and F and atomic number of H is average of atomic number of G and I. Atomic numbers of B, E and H are 7, 15 and 83 respectively and atomic numbers of C, I and F are greater than A, G and D respectively.

9. The incorrect order is :

(A) F > E : Second ionisation energy

(B) B > C : $Z_{eff}$ on valence shell

(C) I > H : First ionisation energy

(D) C > F > E : Electronegativity

10. The correct statement is :

(A) +5 oxidation state of H is more stable than its +3 oxidation state.

(B) $G^{2+}$ is better oxidising agent than $G^{4+}$

(C) +3 oxidation state of E is more stable than its +5 oxidation state

(D) Ionisation energy of G is greater than that of 'Tin',

11. Which of the following statement is incorrect?

(A) $B_2C_5$ is acidic in nature

(B) $AC_2$ is acidic in nature

(C) $FC_3$ is basic in nature

(D) $GC_2$ is amphoteric in nature

12. S, T and U are the aqueous chlorides of the elements X, Y and Z respectively. X, Y and Z are in the same period of the periodic table. U gives a white precipitate with NaOH but this white precipitate dissolves as more NaOH is added. When NaOH is added to T, a white precipitate forms which does not dissolve when more base is added. S does not give precipitate with NaOH.

Which of the following statements are correct?

(A) The three elements are metals

(B) The electronegativity decreases from X to Y to Z

(C) X, Y and Z could be sodium, magnesium and aluminium respectively

(D) The first ionization increases from X to Y to Z

13. There are three elements A, B, and C. Their atomic number are $Z_1$, $Z_2$ and $Z_3$ respectively. If $Z_1 - Z_2 = 2$ and $\frac{Z_1 + Z_2}{2} = Z_3 - 2$ and the electronic configuration of element A is $[Ar]3d^6 4s^2$, then correct order of magnetic momentum is/are:

(A) $B^+ > A^{2+} > C^{2+}$ (B) $A^{3+} > B^{2+} > C$

(C) $B > A > C^{2+}$ (D) $B = A^{3+} > C^{3+}$

14. Consider the value of all four quantum number for last electron and spin multiplicity (2s + 1) for given two element X and Y in their ground state:

| | n | l | m | s | \|2s + 1\| |
|---|---|---|---|---|---|
| X : | 2 | 0 | 0 | +1/2 | 1 |
| Y : | 2 | 1 | –1 | –1/2 | 4 |

The according to given information the correct statement is :

(A) The bond angle (H — Y — H) of possible hydride of element Y is less than 109°28′

(B) The possible halide of X has two vacant p-orbitals on its central atom

(C) Magnetic moment of Y is greater than X

(D) X and Y element exhibits only single oxidation state

15. Consider the following values of I.E. (eV) for elements W and X:

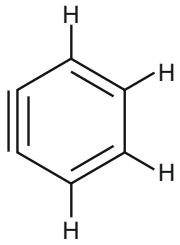
| Element | $I.E._1$ | $I.E._2$ | $I.E._3$ | $I.E._4$ |
|---|---|---|---|---|
| W | 10.5 | 15.5 | 24.9 | 79.8 |
| X | 8 | 14.8 | 78.9 | 105.8 |

Other two element Y and Z have outer electronic configuration $ns^2np^4$ and $ns^2np^5$ respectively. Then according to given information which of the following compound(s) is/are not possible?

(A) $W_2Y_3$ (B) $X_2Y_3$

(C) $WZ_2$ (D) $XZ_2$

16. How many $sp^2$ and *sp*-hybridised carbon atoms are present respectively in the following compound?




(A) 4, 2 (B) 6, 0

(C) 3, 3 (D) 5, 1

17. Give the correct order of initials **T** or **F** for following statements. Use **T** if statement is true and **F** if it is false:

(i) The order of repulsion between different pair of electrons is $l_p - l_p > l_p - b_p > b_p - b_p$

(ii) In general, as the number of lone pair of electrons on central atom increases, value of bond angle from normal bond angle also increases

(iii) The number of lone pair on O in $H_2O$ is 2 while on N in $NH_3$ is 1

(iv) The structures of xenon fluorides and xenon oxyfluorides could not be explained on the basis of VSEPR theory

(A) T T T F (B) T F T F

(C) T F T T (D) T F F F

18. Which of the following overlaps is incorrect (assuming Z-axis is internuclear axis)?

(i) $2p_y + 2p_y \rightarrow \pi$-Bond formation

(ii) $2p_x + 2p_x \rightarrow \sigma$-Bond formation

(iii) $3d_{xy} + 3d_{xy} \rightarrow \pi$-Bond formation

(iv) $2s + 2p_y \rightarrow \pi$-Bond formation

(v) $3d_{xy} + 3d_{xy} \rightarrow \delta$-Bond formation

(vi) $2p_z + 2p_z \rightarrow \sigma$-Bond formation

(A) i, ii, iii (B) iii, vi

(C) ii, v (D) ii, iii, iv

19. Choose the correct code of characteristics for the given order of hybrid orbitals of same atom,

$sp < sp^2 < sp^3$

(i) Electronegativity

(ii) Bond angle between same hybrid orbitals

(iii) Size

(iv) Energy level

(A) (ii), (iii) and (iv)

(B) (iii), (iv)

(C) (ii) and (iv)

(D) (i), (ii), (iii) and (iv)

20. Which of the following is incorrectly match?

| | **Hybridisation** | **Geometry** | **Orbitals use** |
|---|---|---|---|
| (A) | $sp^3d$ | Trigonal bipyramidal | $s + p_x + p_y + p_z + d_{z^2}$ |
| (B) | $sp^3d^3$ | Pentagonal bipyramidal | $s + p_x + p_y + p_z + d_{x^2-y^2} + d_{z^2} + d_{xy}$ |
| (C) | $sp^3d^2$ | Capped octahedral | $s + p_x + p_y + p_z + d_{x^2-y^2} + d_{z^2}$ |
| (D) | $sp^3$ | Tetrahedral | $s + p_x + p_y + p_z$ |

**Paragraph for Q. No. 21**

According to VBT any covalent bond will be formed by overlapping of atomic orbitals of bonded atoms provided atomic orbitals must be half-filled and electrons be in opposite spin. According to type of overlapping covalent bonds can be classified as (a) $\sigma$-bond (b) $\pi$-bond (C) $\delta$-bond.

21. If $F_2C_1 = C_2$ part of $F_2C_1 = C_2 = C_3 = C_4F_2$ lies in yz-plane, then incorrect statement is

(A) Nodal plane of $\pi$-bond between $C_1$ and $C_2$ lies in yz-plane, formed by sideways overlapping of $p_x$-orbitals

(B) Nodal plane of $\pi$-bond between $C_2$ and $C_3$ lies in xz-plane, formed by sideways overlapping of $p_y$-orbitals

(C) Nodal plane of $\pi$-bond between $C_3$ and $C_4$ lies in yz-plane, formed by sideways overlapping of $p_y$-orbitals

(D) Nodal plane of $\pi$-bond between $C_2$ and $C_3$ lies in xy-plane, formed by sideways overlapping of $p_z$-orbitals

22. Consider the following molecule :

$$\underset{(1)}{H_2C} = \underset{(2)}{C} = \underset{(3)}{C} = \underset{(4)}{C} = \underset{(5)}{CF_2}$$

If hybridization of $C_{(1)}$ carbon atom is $sp^2(s + p_y + p_z)$ and hybridization of $C_{(4)}$ carbon atom is $sp(s + p_z)$. Then according to given information the **correct** statement(s) is/are:

(A) Nodal plane of $\pi$-bond between $C_{(2)}$ and $C_{(3)}$ lies in xz-plane, formed by sideways overlapping of $p_y$-orbitals

(B) Nodal plane of $\pi$-bond between $C_{(3)}$ and $C_{(4)}$ lies in yz-plane, formed by side ways overlapping of $p_x$-orbitals

(C) The orbitals involve in hybridization of $C_{(5)}$ carbon atom are $s + p_x + p_z$

(D) Nodal plane of $\pi$-bond between $C_{(1)}$ and $C_{(2)}$ lies in yz-plane, formed by side ways overlapping of $p_y$-orbitals

23. Consider the following two molecules and according to the given information select correct statement(s) about $AX_2$ and $AY_2$ :

Where A : $16^{th}$ group of $3^{rd}$ period element

X : More electronegative than (A) and same group number of (A)

Y : Less atomic size than (A) and same period number of (A)

(A) The hybridization of central atoms are different in both compounds

(B) The shape of both molecules are same

(C) Both compounds are planar

(D) The X — A — X bond angle is less than Y — A — Y bond angle

24. Which of the following combination of bond pair (b.p.) and lone pair (l.p.) give same shape?

(i) 3 b.p. + 1 l.p.

(ii) 2 b.p. + 2 l.p.

(iii) 2 b.p. + 1 l.p.

(iv) 2 b.p. + 0 l.p.

(v) 3 b.p. + 2 l.p.

(vi) 2 b.p. + 3 l.p.

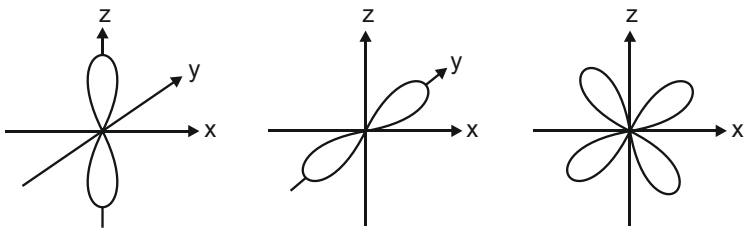
(A) (ii) and (iii)

(B) (iv) and (v)

(C) (iv) and (vi)

(D) (iii) and (vi)

25. Consider the following three orbitals :




(i) (ii) (iii)

Correct statement(s) regarding given information is/are:

(A) Orbitals (i) and (ii) can never form any type of covalent bond

(B) If internuclear axis is x, then combination of (ii) and (iii) orbitals can form $\pi$-bond

(C) Orbital (iii) can form $\delta$-bond with other orbital having same orientation of lobes

(D) If internuclear axis is x, then combination of (i) and (iii) orbitals can form $\pi$-bond

26.

| **Column-I** (Type of bond formed) | **Column-II** [Combining orbitals, (internuclear axis)] |
|---|---|
| (A) $\pi$-bond | (P) $d_{yz} + p_y$, (z) |
| (B) $\sigma$-bond | (Q) $s + p_x$, (y) |
| (C) $\delta$-bond | (R) $d_{yz} + d_{yz}$, (x) |
| (D) Non-bonding | (S) s + s, (z) |
| | (T) $s + d_{xy}$, (y) |

27. Consider the following five groups (According to modern periodic table) of elements with their increasing order of atomic numbers :

Group 1 $\rightarrow$ A, B, C, D, E

Group 2 $\rightarrow$ F, G, H, I, J

Group 13 $\rightarrow$ K, L, M, N, O

Group 15 $\rightarrow$ P, Q, R, S, T

Group 17 $\rightarrow$ U, V, W, X, Y

If first and last element of each group belongs to 2nd and 6th period respectively and Z represents to carbonate ion $\left(CO_3^{2-}\right)$ then consider the following orders.

(i) $O^+ > H^{2+}$ ; Polarising power

(ii) $T^{3+} > S^{3+} > R^{3+}$; Stability of cation

(iii) $U^-(aq) > V^-(aq) > W^-(aq) > X^-(aq)$; size

(iv) $JV_2 < IV_2 < GV_2 < LV_3$; Covalent character

(v) GZ > IZ > JZ; Thermal stability

(vi) AV > BV > CV > DV > EV; Thermal stability

(vii) $C_3P > B_3P > A_3P$; Lattice energy

(viii) $KU_3 < KV_3 < KW_3 < KX_3$; Melting point

Then calculate value of $|p - q|^2$, here p and q are correct and incorrect orders in the given eight orders respectively.

28. Consider the following orbitals:

(i) $3p_x$

(ii) $4d_{z^2}$

(iii) $3d_{x^2-y^2}$

(iv) $3d_{yz}$

Then calculate value of x + y – z here x is total number of gerade orbital and y is total number of ungerade orbitals and z is total number of axial orbital in given above orbitals.

29. $O_2F_2$ is an unstable yellow orange solid and $H_2O_2$ is a colourless liquid both have O — O bond and O — O bond length in $H_2O_2$ and $O_2F_2$ respectively is :

(A) 1.22 Å, 1.48 Å

(B) 1.48 Å, 1.22, Å

(C) 1.22 Å, 1.22 Å

(D) 1.48 Å, 1.48 Å

30. Solubility of alkali metal fluorides increases down the group. Select correct explanation for given statement :

(A) Hydration energy increases and lattice energy decreases down the group

(B) Both energy decrease down the group but decrease in hydration energy is rapid

(C) Both energy decreases down the group but decrease in lattice energy is rapid

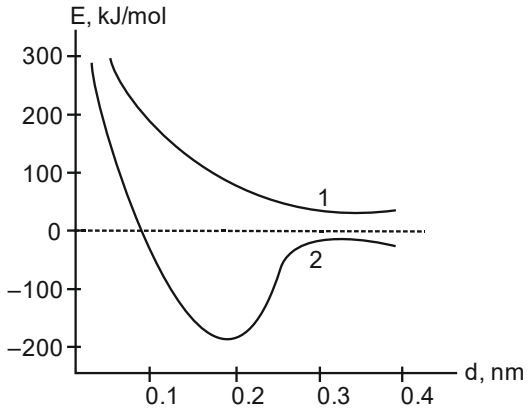
(D) Both energy increase down the group but increase in hydration energy is rapid

31. A molecule may be represented by three structures having energies $E_1$, $E_2$ and $E_3$, respectively. The energies of these structures follow the order $E_3 < E_2 < E_1$, respectively. If the experimental bond energy of the molecule is $E_0$, the resonance energy is :

(A) $(E_1 + E_2 + E_3) - E_0$ (B) $E_0 - E_3$

(C) $E_0 - E_1$ (D) $E_0 - E_2$

32. Consider the given figure showing the formation of $H_2^+$ ion depending on internuclear distance *versus* potential energy of the system.




Which is correct statement ?

(A) Curve-1 represents the most stable state of the system for $H_2^+$ ion

(B) Curve-2 represents the most stable state of the system for $H_2^+$ ion

(C) Curve-1 indicates that the molecular hydrogen ion is formed

(D) Curve-2 represents the energy level of the antibonding region

33. The incorrect statement(s) regarding $\dot{C}X_3$ species is :

(A) If electronegative of surrounding element X is less than 2.5, then central carbon atom used almost 33% s-character in their hybrid bonding orbitals

(B) If electronegativity of surrounding element X is greater than 2.5, then central carbon atom used almost 25% s-character in their hybrid bonding orbitals

(C) If X is F, then species should be polar and pyramidal

(D) If X is H, then species should be polar and planar

34. The incorrect statement about carbene ($CH_2$) is

(A) In singlet carbene, carbon is $sp^2$-hybridized

(B) Triplet carbene is less stable than singlet carbene

(C) Stability order of singlet halocarbenes is CHF > CHCl > CHBr

(D) Triplet Carbene is more stable than singlet carbene

35. Consider the following reaction :

$MX_4 + X'_2 \longrightarrow MX_4X'_2$

If atomic number of M is 52 and X and X′ are halogens and X′ is more electronegative than X. Then choose correct statement regarding given information:

(A) Both X′ atoms occupy axial positions which are formed by overlapping of p and d-orbitals only

(B) All M — X bond lengths are identical in both $MX_4$ and $MX_4\ X'_2$ compounds

(C) Central atom M does not use anyone valence non-axial set of d-orbital in hybridization of final product

(D) Hybridization of central atom M remains same in both reactant and final product

36. Consider the following information (X = F or Cl)

| Molecule | P — X (axial) bond length | P — X (Equatorial) bond length |
|---|---|---|
| $PF_5$ | a | b |
| $PF_4CH_3$ | c | d |
| $PF_3(CH_3)_2$ | e | f |
| $PCl_5$ | g | h |

According to given information choose the incorrect order of bond length:

(A) g > a > d > b (B) g > e > f > b

(C) f > d > a > b (D) c > f > d > b

37. Select the incorrect statement about $N_2F_4$ and $N_2H_4$ :

(i) In $N_2F_4$, d-orbitals are contracted by electronegative fluorine atoms, but d-orbital contraction is not possible by H-atom in $N_2H_4$

(ii) The N — N bond energy in $N_2F_4$ is more than N — N bond energy in $N_2H_4$

(iii) The N — N bond length in $N_2F_4$ is more than that of in $N_2H_4$

(iv) The N — N bond length in $N_2F_4$ is less than that of in $N_2H_4$

Choose the correct code :

(A) i, ii and iii (B) i and iii

(C) ii and iv (D) ii and iii

38. Cis-butene dioic acid $\xrightleftharpoons{Ka_1(-H^+)} X_1^- \xrightleftharpoons{Ka_2(-H^+)} X_2^{2-}$

Trans-butene dioic acid $\xrightleftharpoons{Ka'_1(-H^+)} Y_1^- \xrightleftharpoons{Ka'_2(-H^+)} Y_2^{2-}$

The incorrect statement regarding above information is :

(A) $X_2^{2-}$ species is more basic than $Y_2^{2-}$ species

(B) $X_1^-$ species is more basic than $Y_1^-$ species

(C) $K_{a_1}$ is greater than $K'_{a_1}$

(D) $K'_{a_2}$ is greater than $K_{a_2}$

39. The incorrect statement regarding $O(SiH_3)_2$ and $OCl_2$ molecule is/are :

(A) The strength of back bonding is more in $O(SiH_3)_2$ molecule than $OCl_2$ molecule

(B) Si — O — Si bond angle in $O(SiH_3)_2$ is greater than Cl — O — Cl bond angle in $OCl_2$

(C) The nature of back bond in both molecules is $2p_\pi$-$3d_\pi$

(D) Hybridisation of central O-atom in both molecules is same

40. Hybridisation of central atom does not always change due to back bonding. This statement is valid for which of the following compounds?

(i) $CCl_3^-$

(ii) $BF_3$

(iii) $(SiH_3)_2O$

(iv) $N(SiH_3)_3$

(A) (i), (ii)

(B) (i), (iii)

(C) (ii), (iii)

(D) All

41. The incorrect statement regarding molecular orbital(s) is

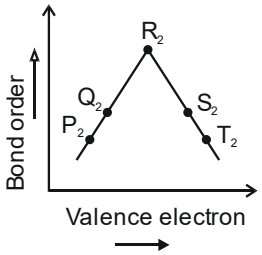
(A) If there is a nodal plane perpendicular to the internuclear axis and lying between the nuclei of bonded atoms then corresponding orbitals is antibonding M.O.

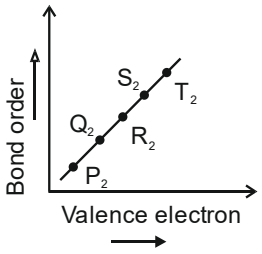
(B) If a nodal plane lies in the inter-nuclear axis, then corresponding orbitals is pi($\pi$) bonding M.O

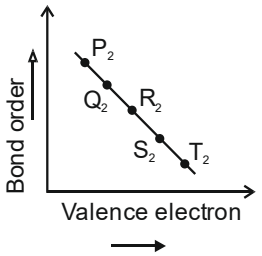
(C) The $\sigma$-bonding molecular orbital does not contain nodal planes containing the internuclear axis

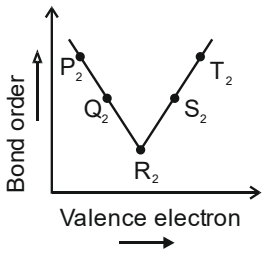
(D) The $\delta$-bonding molecular orbital possesses three nodal planes containing the internuclear axis

42. If P to T are second period p-block elements then which of the following graph show correct relation between valence electrons in $P_2$ to $T_2$ (corresponding molecules) and their bond order?

(A)



(B)



(C)


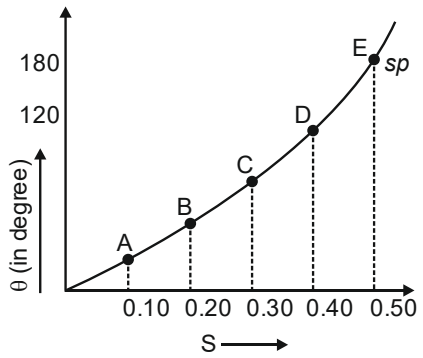

(D)


**Paragraph for Q. Nos. 43 to 45**

| Compound, $MX_n$ type (n = 2 or 3 or 4) | Value of cosθ (θ = bond angle between equivalent hybrid orbitals) |
|---|---|
| P | –0.241 |
| Q | –0.292 |
| R | –0.5 |
| S | –0.325 |
| T | –0.469 |

In all expected compounds each central atom only uses its s and p-orbitals in hybridization. The relationship between bond angle θ and decimal fraction of s and p-character present in the equivalent hybrid orbitals is given by :

$\cos\theta = \frac{S}{S-1} = \frac{P-1}{P}, S =$ decimal fraction of s-character in the equivalent hybrid orbital P = decimal fraction of p-character in the equivalent hybrid orbital.




43. The correct order of % p-character in bond pairs of central atoms in the following compounds is

(A) P > T > S > Q > R

(B) S > R > T > P > Q

(C) P > Q > S > R > T

(D) P > Q > S > T > R

44. If the value of n is 2 for compound T, then number of lone pair present at central atom of compound T will be

(A) 0 (B) 1

(C) 2 (D) 3

45. The incorrect statement is

(A) The ratio of % p-character to % s-character is less than four, for the bond pair of central atom of compound S

(B) Central atom uses three hybrid orbitals to form compound R

(C) Central atom uses four hybrid orbitals to form compound S

(D) There are three compounds present between point C to E, according to % s-character in bond pair of central atom

46. Two compounds $PX_2Y_3$ and $PX_3Y_2$. (Where P = Phosphorous atom and X, Y = monovalent atoms). If all X atoms are replaced by Z atoms and electronegativity order is X > Y > Z.

Then incorrect statement(s) is/are

(A) The dipole moment of product obtained from $PX_2Y_3$ is non-zero

(B) The dipole moment of product obtained from $PX_2Y_3$ is zero

(C) The dipole moment of product obtained from $PX_3Y_2$ is zero

(D) The dipole moment of product obtained from $PX_3Y_2$ is non-zero

47. Consider the following three compounds (i) $AX_{2n}^{n-}$, (ii) $AX_{3n}$ and (iii) $AX_{4n}^{n+}$, where central atom A is 15$^{th}$ group element and their maximum covalency is 3n. If total number of proton in surrounding atom X is n and value of n is one, then calculate value of "$a^3 + b^2 + c$". (Where a, b and c are total number of lone pair at central atom in compound (i), (ii) and (iii) respectively).

48. Consider the following combination of atomic orbitals

| Combinding orbitals (Internuclear axis) | Combinding orbitals (Internuclear axis) |
|---|---|
| (i) $s + p_x(x)$ | (ii) $d_{xy} + d_{xy}(x)$ |
| (iii) $d_{yz} + p_z(z)$ | (iv) $s + s(z)$ |
| (v) $d_{yz} + d_{yz}(x)$ | (vi) $p_y + p_y(y)$ |
| (vii) $d_{z^2} + d_{z^2}(z)$ | (viii) $d_{xy} + d_{xy}(z)$ |
| (ix) $p_x + p_x(y)$ | (x) $s + p_z(x)$ |

Then calculate value of $a^2 + b^2 + 2cd$. (where a = $\sigma$ M. O.; b = $\pi$ M.O.; c = $\delta$ M.O.; d = non-bonding M.O.)

49. Consider the following table

| Compounds (X are monovalent surrounding atoms) | | Central atoms (A to D) belong to group | Characteristics of compounds | Number of lone pair(s) at central atom |
|---|---|---|---|---|
| (i) | $AX_{n_1}$ | 16 | Planar and polar | $m_1$ |
| (ii) | $BX_{n_2}$ | 15 | Trigonal pyramidal | $m_2$ |
| (iii) | $CX_{n_3}$ | 14 | Zero dipole moment | $m_3$ |
| (iv) | $DX_{n_4}$ | 13 | All X — D — X bond angle are 120° | $m_4$ |

Then calculate value of expression $\left|\frac{n_1+n_2+n_3+n_4}{m_1+m_2+m_3+m_4}\right|^2$

50. Consider the following oxy anions

$PO_4^{3-}, P_2O_6^{4-}, SO_4^{2-}, MnO_4^-, CrO_4^{2-}, S_2O_5^{2-}, S_2O_7^{2-}$

and find the value of R + Q – P

Where

P = Number of oxy anions having three equivalent

X — O bond per central atom

Q = Number of oxy anions having two equivalent

X — O bonds per central atom

R = Number of oxy anions having four equivalent

X — O bonds per central atom

51. Consider the following complex : $[Co(NH_3)_5CO_3]ClO_4$

The coordination number, oxidation number, no. of d-electrons and number of unpaired d-electrons on the metal are respectively:

(A) 6, 2, 7, 3 (B) 7, 2, 7, 1

(C) 5, 3, 6, 4 (D) 6, 3, 6, 0

52. In the isoelectronic series of metal carbonyl, the CO bond strength is expected to increase in the order

(A) $[Mn(CO)_6]^+ < [Cr(CO)_6] < [V(CO)_6]^-$

(B) $[V(CO)_6]^- [Cr(CO)_6] < [Mn(CO)_6]^+$

(C) $[V(CO)_6]^- < [Mn(CO)_6]^+ < [Cr(CO)_6]$

(D) $[Cr(CO)_6] < [Mn(CO)_6]^+ < [V(CO)_6]^-$

53. Complexes formed in the following methods are

(i) Mond's process for purification of nickel

(ii) Removal of unreacted AgBr from photographic plate

(iii) Removal of lead poisoning from the body

| | I | II | III |
|---|---|---|---|
| (A) | $Ni(CO)_4$ | $[Ag(CN)_2]^-$ | $[Pb\ (EDTA)]^{2-}$ |
| (B) | $Ni(CO)_4$ | $[Ag(S_2O_3)_2]^{3-}$ | $[Pb\ (EDTA)]^{2-}$ |
| (C) | $Ni(CO)_6$ | $[Ag(S_2O_3)_2]^{3-}$ | $[Pb\ (EDTA)]^{4-}$ |
| (D) | $Ni(CO)_6$ | $[Ag(S_2O_3)]^-$ | $[Pb\ (EDTA)]^{2-}$ |

54. Which of the following order of CFSE is incorrect?

(A) $[Co(en)_3]^{3+} > [Co(NH_3)_6]^{3+} > [Co(H_2O)_6]^{3+}$

(B) $[PtCl_4]^{2-} > [PdCl_4]^{2-} > [NiCl_4]^{2-}$

(C) $[Ni(DMG)_2] < [Ni(en)_2]^{2+}$

(D) $[Co(ox)_3]^{3-} < [Co(en)_3]^{3+}$

55. The IR stretching frequencies of free CO, and CO in $[V(CO)_6]^-$, $[Cr(CO)_6]^-$ and $[Mn(CO)_6]^-$ are 2143 $cm^{-1}$, 1860 $cm^{-1}$, 2000 $cm^{-1}$ and 2090 $cm^{-1}$, respectively. Then correct statement about metal carbonyls is

(A) C — O bond is strongest in the cation and weakest in the anion

(B) C — O bond is weakest in the cation and strongest in the anion

(C) C — O bond is longer in the cation than in the anion

(D) M — C pi bonding is higher in the cation

56. Complex compound(s) having even number of space (stereo) isomers is/are:

(where AA-symmetrical bidentate ligand and a, b, c, d, e-monodentate ligands)

(A) $[M(AA)_2b_2]^{n\pm}$ (B) $[Ma_3b_3]^{n\pm}$

(C) $[Ma_3bcd]^{n\pm}$ (D) $[Ma_2bcde]^{n\pm}$

57. Which of the following statement is true?

(A) In $[PtCl_2(NH_3)_2]^{2+}$ the cis form is optically inactive while trans form is optically active

(B) In $[Fe(C_2O_4)_3]^{3-}$, geometrical isomerism does not exist while optical isomerism exists

(C) In Mabcd, square planar complexes show both optical as well as geometrical isomerism

(D) In Mabcd tetrahedral complex, optical isomerism cannot be observed

58. Choose the correct code regarding, possible number of geometrical isomers exhibited by following complexes

(i) $[CrCl_2(NO_2)_2(NH_3)_2]^-$

(ii) $[Co(NO_2)_3(NH_3)_3]$

(iii) $[PtCl(NO_2)(NH_3)(py)]$

(iv) [PtBrCl(en)]

| | I | II | III | IV |
|---|---|---|---|---|
| (A) | 4 | 2 | 3 | 1 |
| (B) | 5 | 2 | 4 | 0 |
| (C) | 3 | 2 | 0 | 1 |
| (D) | 5 | 2 | 3 | 1 |

59. Consider the following two reactions

$$Cd^{2+}(aq.) + 4CH_3NH_2 \xrightarrow{K_1} A,\ \Delta G_1^\circ$$

$$Cd^{2+}(aq.) + 2H_2NCH_2CH_2NH_2 \xrightarrow{K_2} B,\ \Delta G_2^\circ$$

According to given information the correct statement(s) is/are

(A) $\Delta G_2^\circ$ is more negative than $\Delta G_1^\circ$

(B) Compound A is optically inactive

(C) Compound 'B' is optically active

(D) Formation constant $K_2$ is greater than formation constant $K_1$

60. Consider the following two carbonyl compounds

(i) $[Tc(CO)_6]^+$ and (ii) $[Nb(CO)_6]^-$

Select incorrect statement(s) for given carbonyl compounds.

(A) $[Tc(CO)_6]^+$ acts as reducing agent and $[Nb(CO)_6]^-$ acts as oxidizing agent

(B) $[Nb(CO)_6]^-$ acts as reducing agent and $[Tc(CO)_6]^+$ acts as oxidizing agent

(C) Nb — C bond order in $[Nb(CO)_6]^-$ is greater than Tc — C bond order in $[Tc(CO)_6]^+$

(D) CO bond order is greater in $[Nb(CO)_6]^-$ than in $[Tc(CO)_6]$+

61. Consider the following reactions of complex compounds A, B and C

(i) $\underset{\text{(Compound A)}}{CoCl_2Br \cdot 5NH_3} + \text{excess } Ag^+(aq.) \longrightarrow 1AgCl(s)$

(ii) $\underset{\text{(Compound B)}}{CoCl_2Br \cdot 5NH_3} + \text{excess } Ag^+(aq.) \longrightarrow 2AgCl(s)$

(iii) $\underset{\text{(Compound C)}}{CoCl_2Br \cdot 4NH_3} + \text{excess } Ag^+(aq.) \longrightarrow 1AgCl(s)$

Then according to the given information the correct statement(s) is/are

(A) Compounds (A) and (B) are ionisation isomers

(B) Molar conductivity of compounds (A) and (B) are almost same

(C) Compounds (A), (B) and (C) do not exhibit geometrical isomerism

(D) Order of CFSE values : $\Delta_0(A) > \Delta_0(B) > \Delta_0(C)$

62. Select correct statement(s) regarding given complexes.

(A) $[Fe(CO)_5]$, the orbitals used for hybridization in Fe atom are s, $p_x$, $p_y$, $p_z$, $d_{z^2}$ and it is high spin complex

(B) $[Pt(NH_3)_2Cl_2]$, the orbitals used for hybridization on Pt atom are s, $p_x$, $p_y$, $d_{x^2-y^2}$ and it is low spin complex

(C) $[Cr(H_2O)_6]^{3+}$, the orbitals used for hybridization in Cr atom are s, $p_x$, $p_y$, $p_z$, $d_{z^2}, d_{x^2-y^2}$ and it is high spin complex

(D) $Ni(CO)_4$, the orbitals used for hybridization in Ni atom are, s, $p_x$, $p_y$, $p_z$, and it is low spin complex

63. A d-block element forms octahedral complex but its magnetic moment remains same either in strong field or in weak field ligand. Which of the following is/are correct?

(A) Element always forms colourless compound

(B) Number of electrons in $t_{2g}$ orbitals are higher than in $e_g$, orbitals

(C) It can have either $d^3$ or $d^8$ configuration

(D) It can have either $d^7$ or $d^8$ configuration

64. Consider the following complex compounds

(i) $[Cu(NH_3)_4)\ [Cu(NO_2)_4]$

(ii) $[Cr(py)_6)\ [Cr(SCN)_6]$

(iii) $[Co(NH_3)_5\ (NO_2)\ ]\ [Pt(SCN)_4]$

If $x_1$, $x_2$, $x_3$ are more possible coordination isomers of given complex compounds respectively, then calculate the value of $x_1^2 + x_2^2 + x_3^2$.

65. Consider the following ligands $NH_2^-$, $acac^-$, $OH^-$, $gly^-$, $O_2^-$, o-Phen, $Hdmg^-$, $NO_2^-$, $CO_3^{2-}$, $Cl^-$, $CH_3COO^-$, en, $SO_4^{2-}$. Then calculate the value of "P + Q – R – S".

Here

P : Total numbers of ligands which act as bridging as well as monodentate only

Q : Total number of flexidentate ligands

R : Total number of bidentate ligands only

S : Total number of unsymmetrical bidentate ligands

66. Consider the following reactions at 1000°C

(i) $Zn(s) + \frac{1}{2}O_2(g) \xrightarrow{\Delta} ZnO(g);\ \Delta G° = -360\ kJ\ mol^{-1}$

(ii) $C(s) + \frac{1}{2}O_2(g) \xrightarrow{\Delta} CO(g);\ \Delta G° = -460\ kJ\ mol^{-1}$

and choose the correct statement at 1000°C.

(A) ZnO is more stable than CO

(B) ZnO can be reduced to Zn by C

(C) ZnO and CO are formed at equal rate

(D) ZnO can not be reduced to Zn by C

67. The metal X is prepared by the electrolysis of fused chloride. It reacts with hydrogen to form a colourless solid from which hydrogen gas is released on treatment with water. The metal is

(A) Al (B) Ca

(C) Cu (D) Zn

68. $XCl_2(\text{excess}) + YCl_2 \longrightarrow XCl_4 + Y\downarrow$

$YO \xrightarrow[>400°]{\Delta} \frac{1}{2}O_2 + Y$, Ore of Y would be

(A) Siderite

(B) Cinnabar

(C) Malachite

(D) Horn silver

69. Consider the following metallurgical processes.

(i) Heating impure metal with CO and distilling the resulting volatile carbonyl (b.p. 43°C) and finally decomposition at 150°–200°C to get the pure metal

(ii) Heating the sulphide ore in air until a part is converted to oxide and then further heating in the absence of air to let the oxide react with unchanged metal sulphide

(iii) Electrolysis of the molten electrolyte containing approximately equal amounts of the metal chloride and NaCl to obtained the metal

"The processes used for obtaining magnesium, nickel and copper are respectively :

(A) (i), (ii) and (iii)

(B) (ii), (iii) and (i)

(C) (iii), (i) and (ii)

(D) (ii), (i) and (iii)

70. In the extraction of aluminium

Process X: applied for red bauxite to remove iron oxide (chief impurity)

Process Y: (Serpeck's process) : applied for white bauxite to remove Z (chief impurity) then, process X and impurity Z are :

(A) X = Hall and Heroult's process and Y = $SiO_2$

(B) X = Bayer's process and Y = $SiO_2$

(C) X = Serpeck's process and Y = iron oxide

(D) X = Bayer's process and Y = iron oxide

71. $FeCr_2O_4$ (chromite) is converted to Cr by following steps

$$\text{Chromite} \xrightarrow{i} Na_2CrO_4 \xrightarrow{ii} Cr_2O_3 \xrightarrow{iii} Cr$$

Reagents in i, ii and iii step might be

| | **i-Step** | **ii-Step** | **iii-Step** |
|---|---|---|---|
| (A) | $Na_2CO_3$/air, $\Delta$ | C | C |
| (B) | NaOH/air, $\Delta$ | C, $\Delta$ | Al, $\Delta$ |
| (C) | $Na_2CO_3$/air, $\Delta$ | C, $\Delta$ | C, $\Delta$ |
| (D) | Conc. $H_2SO_4$, $\Delta$ | $NH_4Cl$, $\Delta$ | C, $\Delta$ |

72. Blister copper is refined by stirring molten impure metal with green logs of wood because such a wood liberates hydrocarbon gases (like $CH_4$). This process X is called ..... and the metal contains impurities of Y is .....

(A) X = cupellation, Y = $CuO_2$

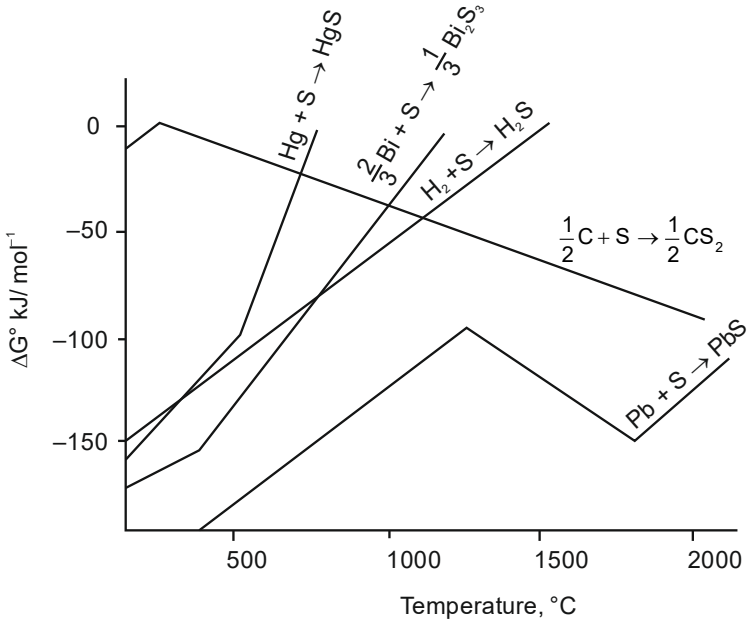
(B) X = polling, Y = $Cu_2O$

(C) X = polling, Y = CuO

(D) X = cupellation, Y = CuO

**Paragraph for Q. Nos. 73 to 75**

The Ellingham diagram for a number of metallic sulphides is shown below.




73. Formation of which of the sulphides is most spontaneous?

(A) HgS (B) $Bi_2S_3$

(C) PbS (D) $CS_2$

74. Which sulphide occurs to minimum extent in nature?

(A) HgS (B) $H_2S$

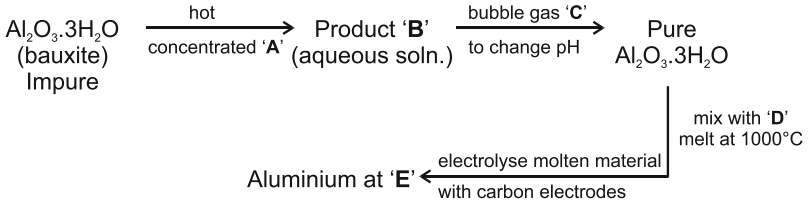
(C) $Bi_2S_3$ (D) $CS_2$

75. Which of the following sulphides can not be reduced to metal by $H_2$ at about 1000°C?

(A) HgS (B) PbS

(C) $Bi_2S_3$ (D) All of these

76. In the extraction of aluminium metal, one of the process is summarised as follows




Which of the following entries correctly summarises reagents, electrodes and products of the process?

| | **A** | **B** | **C** | **D** | **E** |
|---|---|---|---|---|---|
| (A) | NaOH | $Al^{3+}$ | HF | $Na_3AlF_6$ | Cathode |
| (B) | NaOH | $NaAlO_2$ | $CO_2$ | NaF | Anode |
| (C) | $H_2SO_4$ | $Al_2(SO_4)_3$ | $NH_3$ | $Na_3AlF_6$ | Cathode |
| (D) | NaOH | $NaAlO_2$ | $CO_2$ | $Na_3AlF_6$ | Cathode |

77. Column-I and Column-II contains four entries each. Entries of Column-I are to be matched with some entries of Column-II. Each entry of column-I may have the matching with one or more than one entries of Column-II.

| **Column-I (Main ore of metal)** | **Column-II (Process involved in commercial extraction pure metal)** |
|---|---|
| (A) Cinnabar | (P) Froth floatation method |
| (B) Chalcopyrite | (Q) Roasting |
| (C) Bauxite | (R) Distillation |
| (D) Argentite | (S) Leaching |
| | (T) Calcination |

78. When a concentrated solution of ammonia is saturated with sodium chloride in the presence of pieces of dry ice, a water cloud forms. This is due to the

(A) Precipitation of sodium carbonate from the reaction mixture

(B) Precipitation of sodium hydrogen carbonate from the reaction mixture

(C) Precipitation of ammonium hydrogen carbonate from the mixture

(D) Precipitation of ammonium carbonate

79. A compound (A) is used in preparation of washing soda to recover ammonia in Solvay's process. When $CO_2$ is bubbled through an aqueous solution of (A), the solution turns milky. It is used in white washing due to disinfectant nature. What is the chemical formula of A?

(A) $Ca(HCO_3)_2$ (B) CaO

(C) $Ca(OH)_2$ (D) $CaCO_3$

80. $KO_2 + CO_2 + H_2O \xrightarrow{\text{more } CO_2} [X] + [Y]$

Products [X] and [Y] are respectively

(A) $K_2CO_3$, $O_2$ (B) $KHCO_3$, $O_2$

(C) KOH, $K_2CO_3$ (D) $KHCO_3$, $H_2O$

81. Baking powder used to make cake is a mixture of starch, $NaHCO_3$ and $Ca(H_2PO_4)_2$. The function of $Ca(H_2PO_4)_2$ is

(A) To show down the release of $CO_2$ gas

(B) Being acidic in nature and gives $CO_2$ when moistened with $NaHCO_3$

(C) To act as a filler

(D) None of these

82. $A + H_2O \longrightarrow NaOH$; $A \xrightarrow[400°C]{O_2} B \xrightarrow[\text{at } 25°C]{H_2O} NaOH + O_2$

B is used for oxygenating in submarine. A and B are

(A) $Na_2O_2$ and $Na_2O$ (B) $Na_2O$ and $Na_2O_2$

(C) $Na_2O_2$ and $O_2$ (D) $Na_2O$ and $O_2$

83. Calcium imide on hydrolysis gives gas (B) which on oxidation by bleaching powder gives gas (C). Gas (C) on reaction with magnesium give compound (D) which on hydrolysis gives again gas (B). Identify (B), (C) and (D).

(A) $NH_3$, $N_2$, $Mg_3N_2$

(B) $N_2$, $NH_3$, MgNH

(C) $N_2$, $N_2O_5$, $Mg(NO_3)_2$

(D) $NH_3$, $NO_2$, $Mg(NO_2)_2$

84. The more commonly used baking powder contains about 30% $NaHCO_3$, 20% $NaAl(SO_4)_2$, 10% $Ca(H_2PO_4)_2$ and 40% starch. Which of the following statements is/are correct?

(A) $Ca(H_2PO_4)_2$ is acidic and when moistened it reacts with $NaHCO_3$ evolving $CO_2$ gas

(B) $NaAl(SO_4)_2$ slows down the decomposition reaction of $NaHCO_3$ so that $CO_2$ is evolved more slowly

(C) $NaAl(SO_4)_2$ is acidic and when moistened it reacts with $NaHCO_3$ evolving $CO_2$

(D) Both (A) and (B)

**Paragraph for Q. No. 85**

Most metal oxides are thermally stable at temperatures upto 1000°C but the oxide of metals below hydrogen in the electrochemical series decompose fairly easily. Thus HgO and $Ag_2O$ decompose on heating.

85. Correct code for following thermal decomposition reaction(s) evolving gas having equal number of $\sigma$- and $\pi$- bonds is

(i) $BeCO_3 \xrightarrow{\Delta}$

(ii) $ZnSO_4 \xrightarrow[>800°C]{\Delta}$

(iii) $FeSO_4 \xrightarrow{<300°C}$

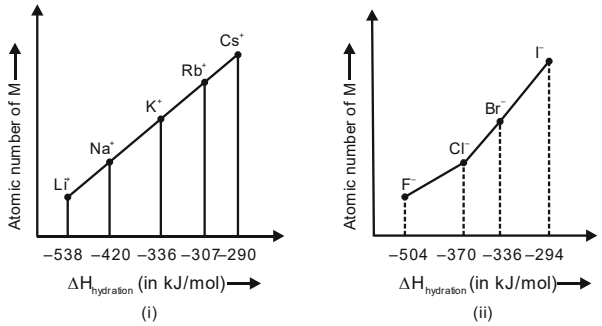
(iv) $(NH_4)_2Cr_2O_7 \xrightarrow{\Delta}$

(A) i, ii (B) i, iii, iv

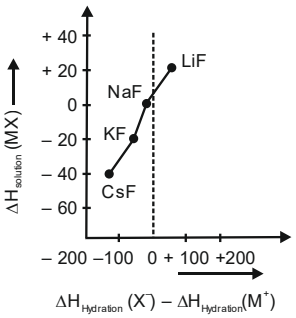
(C) i, ii, iii (D) All of these

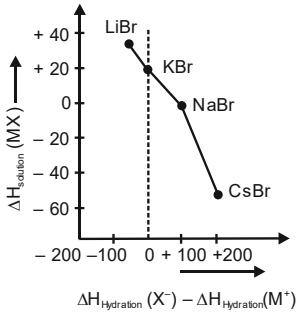
86. Consider the following two graphs between atomic number of element and hydration enthalpies (at 25°C) of corresponding ion.

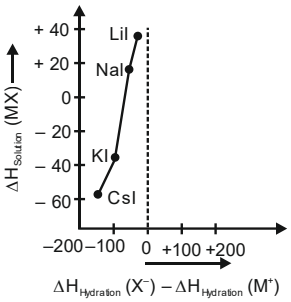
(M = Alkali metal and X = halogens)




Then according to the given information which of the following is/are correct relationship between enthalpy of solution ($\Delta H_{solution}$) of a salt MX in water and difference of the enthalpies of hydration ($\Delta H_{hydration}$) of the constituent ion ($M^+$ and $X^-$)?

(A) 


(B) 


(C) 


(D) None of these

87. The silicate anion in the mineral kinoite is a chain of three $SiO_4$ tetrahedra, that share corners with adjacent tetrahedra. The charge of the silicate anion is:

(A) –4 (B) –8

(C) –6 (D) –2

88. $H_2C_2O_4 \xrightarrow{\Delta} \text{gas (A)} + \text{gas (B)} + \text{liquid (C)}$

Gas (A) burns with a blue flame and is oxidised to gas (B)

$\text{Gas (A)} + Cl_2 \longrightarrow (D) \xrightarrow{NH_3, \Delta} (E)$

A, B, C and E are

(A) $CO_2$ , CO, $H_2O$, $HCONH_2$

(B) CO, $CO_2$, $COCl_2$, $HCONH_2$

(C) CO, $CO_2$, $H_2O$, $NH_2CONH_2$

(D) CO, $CO_2$, $H_2O$, $COCl_2$

89. $BX_3 + NH_3 \xrightarrow{\text{R.T.}} BX_3 \cdot NH_3 + \text{Heat of adduct formation } (\Delta H)$

The numerical value of $\Delta H$ is found ro be maximum for

(A) $BF_3$ (B) $BCl_3$

(C) $BBr_3$ (D) $BI_3$

90. Choose the correct sequence for the geometry of the given molecules.

$Al_2(CH_3)_6$, Borazole, $B_3O_6^{3-}$, $Fe_2Cl_6$, $I_2Cl_6$

['P' stands for planar and 'NP' stands for non-planar]

(A) NP, NP, NP, P, P (B) P, P, NP, NP, P

(C) NP, NP, NP, P, NP (D) NP, P, P, NP, P

91. The correct code for stability, of oxidation states for given cations is

(i) $Pb^{2+} > Pb^{4+}$, $Tl^{+} < Tl^{3+}$

(ii) $Bi^{3+} > Sb^{3+}$, $Sn^{2+} < Sn^{4+}$

(iii) $Pb^{2+} > Pb^{4+}$, $Bi^{3+} > Bi^{5+}$

(iv) $Tl^{3+} < In^{3+}$, $Sn^{2+} > Sn^{4+}$

(v) $Sn^{2+} < Pb^{2+}$, $Sn^{4+} > Pb^{4+}$

(vi) $Sn^{2+} < Pb^{2+}$, $Sn^{4+} < Pb^{4+}$

(A) (v) and (vi) (B) (i), (iii) and (vi)

(C) (iii) and (v) (D) (ii) and (iv)

92. Two oxides of nitrogen, NO and $NO_2$ are allowed to react together at 243 K and form a coloured compound of nitrogen (X). When compound (X) reacts with water to yield another compound of nitrogen (Y). The geometry of the anion of (Y) molecule is

(A) Triangular planar (B) Triangular pyramidal

(C) Tetrahedron (D) Square planar

93. $NH_3 + O_2 \xrightarrow[\Delta]{Pt} A + H_2O$;

$A + O_2 \longrightarrow B$ ;

$B + O_2 + H_2O \longrightarrow C$

A, B and C are

(A) $N_2O$, $NO_2$ and $HNO_3$

(B) NO, $NO_2$ and $HNO_3$

(C) $NO_2$, NO and $HNO_3$

(D) $N_2O$, NO and $HNO_3$

94. Which of the following properties of halogens increase with increasing atomic number?

(i) Ionization energy

(ii) Ionic radius

(iii) Bond energy of the $X_2$ molecule

(iv) Enthalpy of vaporisation

(A) (i), (ii), (iii) (B) (i), (iii)

(C) (ii), (iv) (D) (iv)

95. $Cl_2(g) + Ba(OH)_2 \longrightarrow X(aq.) + BaCl_2 + H_2O$

$X + H_2SO_4 \longrightarrow Y + BaSO_4$

$Y \xrightarrow[\Delta > 365\,K]{\Delta} Z + H_2O + O_2$

Y and Z are respectively

(A) $HClO_4$, $ClO_2$ (B) $HClO_3$, $ClO_2$

(C) $HClO_3$, $ClO_6$ (D) $HClO_4$, $Cl_2O_7$

96. The three elements X, Y and Z with electronic configurations shown below all form hydrides

| Element | Electronic configuration |
|---|---|
| X | $1s^2 2s^2 2p^2$ |
| Y | $1s^2 2s^2 2p^6 3s^1$ |
| Z | $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^5$ |

Which set of properties match correctly with properties of the hydrides of these elements

| | Hydride of X | Hydride of Y | Hydride of Z |
|---|---|---|---|
| (A) | Colourless gas insoluble in $H_2O$ | Silver/grey solid reacts with $H_2O$ to form an alkali | Colourless gas forms a strong acid in $H_2O$ |
| (B) | Colourless liquid, no reaction with $H_2O$ | Silver/grey solid forms $H_2$ and $H_2O$ | Ionic solid with formula ZH |
| (C) | Non-polar compound reacts with $Cl_2$ in light | Silver/grey ionic solid with formula $YH_2$ | Forms when water is added to phosphorus and elemental Z |
| (D) | Colourless gas which burns with air | Silver/grey solid which reacts violently with acids | Colourless, corrosive liquid at STP |

97. Consider the following transformations

(i) $XeF_6 + NaF \longrightarrow Na^+ [XeF_7]^-$

(ii) $2PCl_5(s) \longrightarrow [PCl_4]^+ [PCl_6]^-$

(iii) $[Al(H_2O)_6]^{3+} + H_2O \longrightarrow [Al(H_2O)_5OH]^{2+} + H_3O^+$

Possible transformations are

(A) (i), (ii), (iii) (B) (i), (iii)

(C) (i), (ii) (D) (ii), (iii)

98. Among the following, cyclic species are

(i) $H_5P_3O_{10}$

(ii) $[B_3O_3(OH)_5]^{2-}$

(iii) Trimer of $SO_3(S_3O_9)$

(iv) $P_3N_3Cl_6$

(A) (i), (iii) (B) (i), (iii), (iv)

(C) (ii), (iii), (iv) (D) (i), (ii), (iii), (iv)

**Paragraph for Q. Nos. 99 to 102**

(i) $P + C \text{ (carbon)} + Cl_2 \longrightarrow Q + CO\uparrow$

(ii) $Q + H_2O \longrightarrow R + HCl$

(iii) $BN + H_2O \longrightarrow R + NH_3\uparrow$

(iv) $Q + LiAlH_4 \longrightarrow S + LiCl + AlCl_3$

(v) $S + H_2O \longrightarrow R + H_2\uparrow$

(vi) $S + NaH \longrightarrow T$

**(P, Q, R, S and T do not represent their chemical symbols)**

99. Compound Q has

(i) Zero dipole moment
(ii) A planar trigonal structure
(iii) An electron deficient compound
(iv) A Lewis base

Choose the correct code

(A) (i), (iv)
(B) (i), (ii), (iv)
(C) (i), (ii) (iii)
(D) (i), (ii), (iii), (iv)

100. Compound T is used as a/an

(A) Oxidising agent
(B) Complexing agent
(C) Bleaching agent
(D) Reducing agent

101. Compound S is

(i) An oldd-$e^-$ compound
(ii) (2c – $3e^-$) compound
(iii) A electron deficient compound
(iv) A $sp^2$ hybridized compound

Choose the correct code

(A) (iii)
(B) (i), (iii)
(C) (ii), (iii), (iv)
(D) (i), (ii), (iv)

102. Consider the following reactions

$$A_x + yB_2 \xrightarrow[\text{supply of air}]{\text{limited}} \text{compound P} \xrightarrow[\text{excess air}]{+zB_2} \text{Compound Q}$$

If atomic number of elements A and b are 15 and 8 respectively, then according to the given information the correct statement(s) is/are

(A) (B — A — B) bond angle in compound Q > (B — A — B) bond angle in compound P
(B) (A — B) bond length in compound Q < (A — B) bond length in compound P
(C) Value of x + y + z is 9
(D) Compound P is $P_2O_5$ and compound Q is $P_4O_{10}$

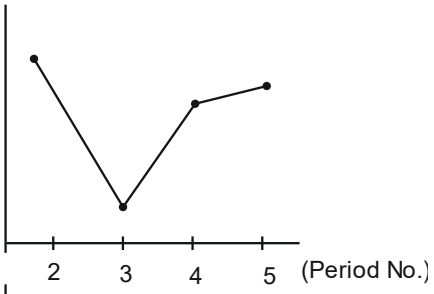
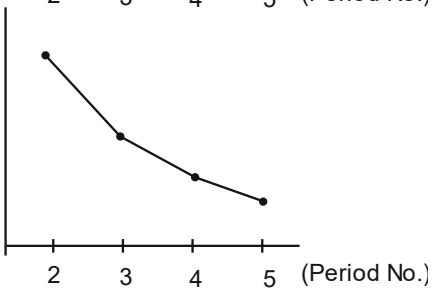
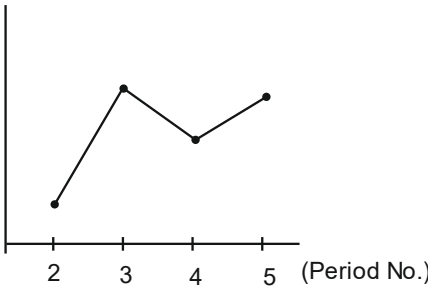
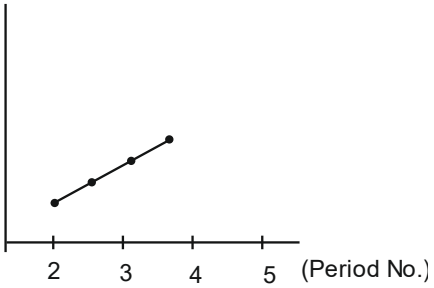
103. Consider the following table

| Compound | Enthalpy of formation (kJ $mol^{-1}$) | Bond angle (H — C.A. — H) | Boiling point (°C) |
|---|---|---|---|
| $H_2O$ | $x_1$ | $y_1$ | $z_1$ |
| $H_2S$ | $x_2$ | $y_2$ | $z_2$ |
| $H_2Se$ | $x_3$ | $y_3$ | $z_3$ |
| $H_2Te$ | $x_4$ | $y_4$ | $z_4$ |

According to given information the correct order is/are

(A) $x_1 > x_2 > x_3 > x_4$
(B) $y_1 > y_2 > y_3 > y_4$
(C) $z_1 > z_4 > z_3 > z_2$
(D) $x_1 > x_4 > x_3 > x_2$

104. Match the Column-I with the graph of Column-II which is most close to the answer.

| Column-I | Column-II |
|---|---|
| (A) Bond energies of the hydra acids of the halogens fluorine, chlorine, bromine, iodine | (P)   |
| (B) Boiling points of the hydrides of the 16 group elements oxygen, sulphur, selenium, tellurium | (Q)   |
| (C) The stability of monochlorides of group 13 elements boron, aluminium, gallium, indium | (R)   |
| (D) Melting points of the dioxides of the group 14 elements carbon, silicon, germanium, tin | (S)   |

105. Consider the structure of $Al_2Me_6$ compound and find the value of $\frac{x-y}{z}$

Where x = Maximum number of atoms that can lie in plane having terminal (Al – Me) bonds.

y = Total number of 3c – $2e^-$ bonds

z = Total number of atoms that are $sp^3$ hybridized

106. Consider the following covalent compounds in their solid state and find the value of expression (X + Y + Z)

$N_2O_5$, $Cl_2O_6$, $PCl_5$, $I_2Cl_6$, $XeF_6$, $PBr_5$

Where X = Total number of compounds in which central atom of cationic or anionic part is $sp^3$ hybridized

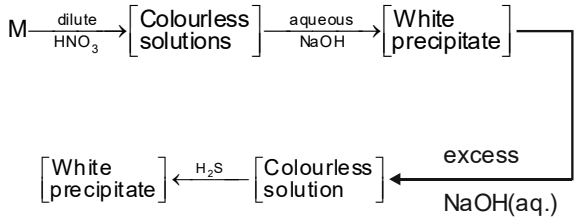
Y = Total number of compounds having 90° bond angle either in cationic or anionic part

Z = Total number of compounds having 109°28' bond angle either in cationic or anionic part

107. When $B_2H_6$ is allowed to react with following Lewis bases, then how many given Lewis bases form adduct through symmetrical cleavage of $B_2H_6$?

$NH_3$, $MeNH_2$, Pyridine, CO, T.H.F, $PH_3$, $PF_3$, $Me_3N$, $Me_2NH$

108. A metal M and its compound can give the following observable changes in a consequence of reactions.




(A) Mg (B) Pb

(C) Zn (D) Sn

109. $AgNO_3 \xrightarrow{\Delta} (W) + (X) + O_2$

(X) + $H_2O \longrightarrow HNO_2 + HNO_3$

(W) + $HNO_3 \longrightarrow Y + NO + H_2O$

(Y) + $Na_2S_2O_3$ (excess) $\longrightarrow$ (Z) + $NaNO_3$

Identify (W) to (Z)

(A) W = Ag X = $N_2O$ Y = $AgNO_3$ Z = $Na_2[Ag(S_2O_3)_2]$

(B) W = $Ag_2O$ X = NO Y = $AgNO_3$ Z = $Na_3[Ag(S_2O_3)_2]$

(C) W = Ag X = $NO_2$ Y = $AgNO_3$ Z = $Na_3[Ag(S_2O_3)_2]$

(D) W = $Ag_2O$ X = $N_2$ Y = $AgNO_3$ Z = $Na[Ag(S_2O_3)_2]$

110. Limestone is present in the blast furnace production of iron in order to

(i) Provide a source of CaO

(ii) Remove some impurities

(iii) Supply $CO_2$

(A) (i), (ii), (iii) (B) (i), (ii)

(C) (ii), (iii) (D) (i) only

111. When aqueous sodium hydroxide is added to an aqueous solution of chromium (III) ions, a green blue precipitate is first formed which re-dissolves to give a green solution. This green colour is due to

(A) $[Cr(H_2O)_6]^{3+}$ (B) $CrO_4^{2+}$

(C) $[Cr(OH)_4]^-$ (D) $[Cr(OH)_3(H_2O)_3]$

112. $FeCr_2O_4 + Na_2CO_3 + O_2 \xrightarrow{\text{Fusion}} [X] \xrightarrow[H_2O]{H^+} [Y] \xrightarrow{H_2O_2/H^+} [Z]$

Which of the following statement is true for the compounds [X], [Y] and [Z]?

(A) In all three compounds, the chromium is in +6 oxidation state

(B) [Z] is a deep blue-violet coloured compound which decomoposes rapidly in aqueous solution into $Cr^{+3}$ and dioxygen

(C) Saturated solution of [Y] gives bright orange compound, chromic anhydride, with concentrated $H_2SO_4$

(D) All of these

113. Consider the following transformation

$$2CuX_2 \xrightarrow{\text{Room temperature}} 2CuX + X_2 \uparrow$$

Then $X^-$ can be

(A) $F^-$, $Br^-$ (B) $Cl^-$, $Br^-$

(C) $CN^-$, $I^-$ (D) $Cl^-$, $F^-$

**Paragraph for Q. No. 114**

Iron (+II) is one of the most important oxidation states, and salts are called ferrous salts. Most of the Fe (+II) salts are pale green and contain $[Fe(H_2O)_6]^{2+}$ ion. Fe(+II) compounds are easily oxidised by air and so are difficult to obtain pure $Fe^{2+}$ from many complexes like $K_4[Fe(CN)_6]$

114. $FeSO_4$ is used in brown ring test for nitrates and nitrites. In this test, a freshly prepared $FeSO_4$ solution is mixed with solution containing $NO^-_2$ or $NO_3^-$ and the conc. $H_2SO_4$ is run down the side of the test tube. If the mixture gets hot or is shaken.

(i) The brown colour disappear

(ii) NO is evolved

(iii) A yellow solution of $Fe_2(SO_4)_3$ is formed

(A) (i), (ii), (iii) correct (B) (i), (iii) correct

(C) (ii), (iii) correct (D) Only (i) correct

115. Consider the following sequence of reactions

$$M^{2+}(aq.) \xrightarrow{NH_4Cl(s)+(NH_4)_2CO_3\,Sol.} Q \downarrow \xrightarrow[\text{Followed by addition of } (NH_4)_2C_2O_4]{CH_3COOH} R \downarrow$$

Which of the following cation can form ppt. Q but does not form ppt. R?

(A) $Mg^{2+}$(aq.) (B) $Ca^{2+}$(aq.)

(C) $Sr^{2+}$(aq.) (D) $Ba^{2+}$(aq.)

116. $M(\text{Salt}) + \text{Dil. HCl} \xrightarrow{\text{Warm}} N \uparrow + P \downarrow$

gas N changes colour of $FeSO_4$ solution into yellow solution then salt M in above reaction is

(A) $BaS_2O_3$ (B) $Ag_2SO_3$

(C) $AgNO_2$ (D) $Pb(NO_3)_2$

117. Sodium salt solution + $AgNO_3$ soln. $\longrightarrow$ Coloured precipitate.

If coloured precipitate is soluble in both dil. $HNO_3$ and excess conc. $NH_3$ solution then which of the following anion is present in the salt solution?

(A) $S^{2-}$(aq.) (B) $I^-$(aq.)

(C) $PO_4^{3-}(aq,)$ (D) $Br^-$(aq.)

118. Fe(Finely powdered) + HCl(dil.) $\longrightarrow$ P + Q $\uparrow$

compound P does not precipitate with

(A) $AgNO_3$ (B) $K_3[Fe(CN)_6]$

(C) KCN (excess) (D) $NH_4Cl + NH_4OH$

119. Bromine is commercially prepared from sea water by displacement reaction

$Cl_2 + 2Br^-(aq.) \longrightarrow 2Cl^-(aq.) + Br_2$

$Br_2$ gas thus formed is dissolved into solution of $Na_2CO_3$ and then pure $Br_2$ is obtained by treatment of the solution with

(A) $Ca(OH)_2$ (B) NaOH

(C) $H_2SO_4$ (D) HI

120. In a closed container there is a mixture of $SO_2$, $CO_2$ and $O_2$ gas. Which sequence of reagent can be helpful to separate them?

(i) Limewater

(ii) Acidified potassium dichromate

(iii) Alkaline pyrogallol

(A) (i), (ii) and (iii) (B) (ii), (i), (iii)

(C) (iii), (ii), (i) (D) (iii), (i), (ii)

121. $Xe + F_2 \xrightarrow{1:20} X \xrightarrow{H_2O} Y \xrightarrow{H_2O} Z \xrightarrow{H_2O} XeO_3$

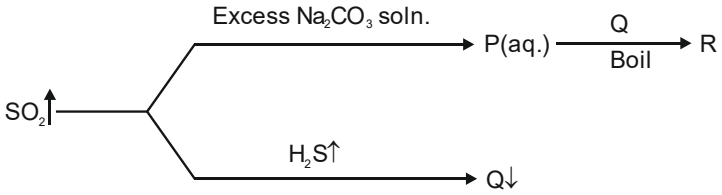
Select correct option(s) for X, Y, Z and given chemical change

(A) X, Y and Z are in same oxidation state

(B) All have equal number of lone pair on central atom

(C) All are non-planar

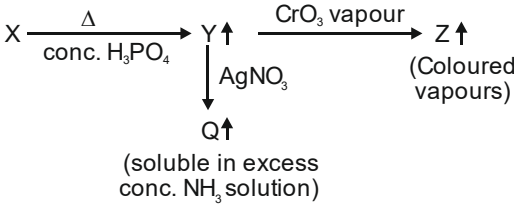
(D) All have equal number of covalent bonds

122. 


Incorrect statement about R is

(A) Antichlor agent (B) Fixing agent in photography

(C) Forms ppt. with $CaCl_2$ solution (D) Reduces $Cu^{2+}(aq,)$ cation

123. 


Which of the following anion cannot be in X?

(A) $F^-$ (B) $Cl^-$

(C) $Br^-$ (D) $I^-$

124. Find the value of expression |x – y| for following compounds.

where,

x = Total number of water insoluble salts.

y = Total number of salts, which can liberate non-polar acidic gas during their complete thermal decomposition,

$BaCO_3$, $PbSO_4$, $AgNO_3$, $CaC_2O_4$, $CsHCO_3$,

$Na_3PO_4$, $CH_3COOAg$, $Mg(OH)_2$, $Pb(NO_3)_2$

125. A reddish pink substance on heating gives off a vapour which condenses on the sides of the test tube and the substance turns blue. It on cooling water is added to the residue it turns to its original colour. The substance is :

(A) Iodine crystals (B) Copper sulphate crystals

(C) Cobalt chloride crystals (D) Zinc oxide

126. A colourless water soluble solid 'X' on heating gives equimolar quantities of Y and Z. Y gives dense white fumes HCl and Z does so with $NH_3$. Y gives brown precipitate with Nessler's reagent and Z gives white precipitate with nitrates of $Ag^+$, $Pb^{2+}$ and $Hg^+$. 'X' is:

(A) $NH_4Cl$ (B) $NH_4NO_3$

(C) $NH_4NO_2$ (D) $FeSO_4$

127. A white crystalline salt imparts a violet colour to Bunsen flame, and with hot concentrated $H_2SO_4$, forms a pungent gas. On treatment with an $AgNO_3$ solution, this gas forms a white precipitate readily soluble in $NH_3$. The white crystalline salt may be

(A) $Na_2SO_4$ (B) KCl

(C) $CaCl_2$ (D) $SrCl_2$

128. A white solid gives a green residue on being subjected to the cobalt nitrate test. On being warmed with concentrated $H_2SO_4$, the solid gives a brown gas, which evolves vigorously on the addition of Cu turnings. The solid may be

(A) $Zn(NO_3)_2$ (B) $Al(NO_3)_2$

(C) $ZnBr_2$ (D) $Mg(NO_3)_2$

129. A compound 'X' on heating gives a colourless gas. The residue is dissolved in water to obtain 'Y'. Excess $CO_2$ is passed through aqueous solution of 'Y' when 'Z' is formed. 'Z' on gentle heating gives back 'X'. The compound 'X' is:

(A) $NaHCO_3$ (B) $Na_2CO_3$

(C) $Ca(HCO_3)_2$ (D) $CaCO_3$

130. Solid KCl, when heated with solid $K_2Cr_2O_7$ and concentrated $H_2SO_4$, gives red vapours (a) that turn NaOH solution yellow (b). The yellow solution, when acidified with acetic acid and treated with lead acetate, gives a yellow precipitate (C). Which of the following is true with respect to a, b and c?

(A) a and b contain $CrO_4^{2-}$

(B) a and c contain $CrO_4^{2-}$

(C) a, b and c contain $CrO_4^{2-}$

(D) a, b and c contain Cr(VI)

131. Choose the correct code by identifying (X), (Y) and (Z) in each case for the changes indicated:

(i) $CrO_2Cl_2 \xrightarrow{KOH} (X) \xrightarrow{\text{conc. } H_2SO_4} (Y) \xrightarrow{AgNO_3} (Z)$

(ii) $CrCl_3(aq.) \xrightarrow[NaOH]{\text{excess}} (X) \xrightarrow[H_2O,\text{ boil}]{Na_2O_2} (Y) \xrightarrow[\text{acetate}]{\text{lead}} (Z)$

(iii) $ZnSO_4(aq.) \xrightarrow{Na_2CO_3} (X) \xrightarrow{\Delta} (Y) \xrightarrow[\text{nitrate, }\Delta]{\text{cobalt}} (Z)$

(iv) $CuCl_2(aq.) \xrightarrow[H_2S]{NH_4OH} (X) \xrightarrow[\Delta]{HNO_3} (Y) \xrightarrow[\text{excess}]{KCN} (Z)$

(A) X = $K_2CrO_4$ Y = $K_2Cr_2O_7$ Z = $Ag_2CrO_4$

(B) X = $[Cr(OH)_4]^-$ Y = $Na_2CrO_4$ Z = $PbCrO_4$

(C) X = $ZnCO_3$ Y = ZnO Z = $CoZnO_2$

(D) X = CuS Y = $Cu(NO_3)_2$ Z = $K_3[Cu(CN)_4]$

132. A sulphate of a metal (A) on heating evolves two gases (B) and (C) and an oxide (D). Gas (B) turns $K_2Cr_2O_7$ paper green while gas (C) forms a trimer in which there is no S-S bond. Compound (D) with conc. HCl forms a Lewis acid (E) which exists in a dimer. Compounds (A), (B), (C), (D) and (E) are respectively

(A) $FeSO_4$ , $SO_2$ , $SO_3$, $Fe_2O_3$, $FeCl_3$

(B) $Al_2(SO_4)_3$, $SO_2$, $SO_3$, $Al_2O_3$, $FeCl_3$

(C) FeS, $SO_2$ , $SO_3$ , $FeSO_4$ , $FeCl_3$

(D) FeS, $SO_2$, $SO_3$, $Fe_2(PO_4)_3$, $FeCl_2$

133. An organic compound (A) on heating produces two gases (B) and (C) and neutral oxide (D) which turns cobalt chloride paper pink. Gas (B) turns lime water milky and produces an acidic solution with water. Gas (C) produces a poisonous gas (E) with chlorine gas, this gas with ammonia gives an organic compound (F) which on further reaction with (D) gives $NH_3$ gas.

Then, compounds (A) and (F) can be found as:

(A) $H_2C_2O_4$ and $NH_2CONH_2$

(B) $CH_3COOH$ and $NH_2CONH_2$

(C) $CHCl_3$ and $CH_3CONH_2$

(D) $CH_3Cl$ and $NH_2COONH_4$

134. A white solid forms Rinmann's green in the charcoal cavity test in an oxidising flame. On treatment with dilute $H_2SO_4$, this solid produces a gas that turns an acidified dichromate paper green and lead acetate paper black. The white solid is

(A) PbS

(B) $ZnSO_3$

(C) ZnS

(D) $Na_2S$

135. $\underset{\text{Imparts violet colour to flame}}{(T)} \xrightarrow{\text{compound (U)} + \text{conc. } H_2SO_4} \underset{\text{Red gas}}{(V)} \xrightarrow{NaOH + AgNO_3} \underset{\text{Red ppt.}}{(W)} \xrightarrow{NH_3 \text{ soln.}} (X)$

$(W) \xrightarrow{\text{dil. HCl}} (Y)$ white ppt.

$(U) \xrightarrow[\Delta]{NaOH} (Z)$ gas (gives white fumes with HCl)

Identify (T) to (Z).

(A) T = $KMnO_4$, U = HCl, V = $Cl_2$, W = $HgI_2$, X = $Hg(NH_2)NO_3$, Y = $Hg_2Cl_2$, Z = $N_2$

(B) T = $K_2Cr_2O_7$, U = $NH_4Cl$, V = $CrO_2Cl_2$, W = $Ag_2CrO_4$, X = $[Ag(NH_3)_2]^+$, Y = AgCl, Z = $NH_3$

(C) T = $K_2CrO_4$, U = KCl, V = $CrO_2Cl_2$, W = $HgI_2$, X = $Na_2CrO_4$, Y = $BaCO_3$, Z = $NH_4Cl$

(D) T = $K_2MnO_4$, U = NaCl, V = $CrO_3$, W = $AgNO_2$, X = $(NH_4)_2CrO_4$, Y = $CaCO_3$, Z = $SO_2$

136. A white powder solid A forms a light green solution with water, which on treatment with potassium hexacyanoferrate(III) gives a blue precipitate. On being strongly heated, A leaves a brown residue and forms a mixture of two gaseous oxides, which turns a dichromate solution green and forms a white precipitate with a $BaCl_2$ solution containing concentrated HCl, 'A' is:

(A) $CuSO_4$ (B) $Fe_2(SO_4)_3$

(C) $FeSO_4$ (D) $Cr_2(SO_4)_3$

137. A white sublimable solid, when boiled with an NaOH solution, gives a colourless gas that turns Nessler's reagent brown. The solid, on being heated with solid $K_2Cr_2O_7$ and concentrated $H_2SO_4$, gives red brown vapours. The white solid can be

(A) $NH_4I$ (B) $NH_4Br$

(C) $NH_4Cl$ (D) $(NH_4)_2SO_4$

138. $\underset{\text{Black}}{X(s)} \xrightarrow{\text{dil. HCl}} Y \uparrow \xrightarrow[NaOH]{[Na_2(Fe(CN))_5(NO)]}$ Purple solution Gas Y has been allowed to react with following species in neutral/acidic medium

(A) $FeCl_3$ (B) $CuSO_4$

(C) $BaCl_2$ (D) $SO_2$

(E) $Cr_2O_7^{2-}$ (F) $CH_3COONa$

(G) $Hg^{2+}$

Then calculate value of (P + Q – R)

P : Number of species which undergoes redox reaction with gas Y

Q : Number of species which give black ppt with gas Y

P : Number of species with which gas Y produce no observable change

❑ ❑ ❑

Book Crunch

# Mathematics

# MATHEMATICS

1. $I_1 = \int \sin^{-1} x\,dx$ and $I_2 = \int \sin^{-1}\sqrt{1-x^2}\,dx$, then

(1) $I_1 = I_2$ (2) $I_2 = \frac{\pi}{2I_1}$

(3) $I_1 + I_2 = \frac{\pi x}{2}$ (4) $I_1 - I_2 = \frac{\pi x}{2}$

2. $\int e^x \left(\frac{1+\sin x\cos x}{\cos^2 x}\right)dx$ is

(1) $e^x \sec^2 x + C$ (2) $e^x \tan x + C$

(3) $e^x \cdot \ln\sec x + C$ (4) None of these

3. If $\int \frac{(x^2-1)}{x\sqrt{x^4+3x^2+1}}dx$ is $m\ln\left[\frac{\{(x^2+1)\}+\sqrt{x^4+3x^2+1}}{x}\right]+C$. Then $m$ is

(1) 1 (2) 2

(3) $\frac{1}{3}$ (4) None of these

4. $\int\left(\frac{dx}{\sin x+\sec x}\right)$ is

(1) $\frac{1}{2\sqrt{3}}\ln\left|\frac{\tan x+\sqrt{x}}{\tan x-\sqrt{3}}\right|+\tan^{-1}(\sin x+\cos x)+C$ (2) $\frac{1}{2\sqrt{3}}\ln\left|\frac{\sin x-\cos x+\sqrt{3}}{-\sin x+\cos x+\sqrt{3}}\right|+\tan^{-1}(\sin x+\cos x)+C$

(3) $\frac{1}{2\sqrt{3}}\ln\left|\frac{\tan x+\sqrt{3}}{\tan x-\sqrt{3}}\right|+\tan^{-1}(\sin x+\cos x)+C$ (4) None of these

5. $\int \frac{dx}{\sqrt{x}\cdot(x^{1/4}+1)^{10}}$ is

(1) $-\frac{4}{8}\left(\frac{1}{(\sqrt[4]{x}+1)^8}\right)+\frac{4}{9}x+C$ (2) $-\frac{4}{8}\frac{1}{(\sqrt[4]{x}+1)^9}+\frac{4}{9(\sqrt[4]{x}+1)^8}+C$

(3) $\frac{4}{9(4\sqrt{x}+1)^9}+C$ (4) None of these

6. $\int \frac{dx}{\sqrt[4]{(x-1)^3(x+2)^5}}$ is

(1) $\frac{3}{4}\left(\frac{x-1}{x+2}\right)^{1/4}$ (2) $\frac{4}{3}\left(\frac{x-1}{x+2}\right)^{1/4}+C$

(3) $\frac{4}{3}\left(\frac{x-1}{x+2}\right)^{-1/4}+C$ (4) None of these

7. The value of $\int \frac{\cos^3 x+\cos^5 x}{\sin^2 x+\sin^4 x}dx$ is

(1) $\sin x-6\tan^{-1}(\sin x)+C$ (2) $\sin x-2(\sin x)^{-1}+C$

(3) $\sin x-2(\sin x)^{-1}+6\tan^{-1}(\sin x)+C$ (4) None of these

8. $\int \frac{e^{(x)^{1/2}}\cos e^{(x)^{1/2}}}{(x)^{1/2}}dx$ is equal to

(1) $2\cos e^{(x)^{1/2}}+C$ (2) $2(\cos e^x)^{1/2}+C$

(3) $2\sin e^{(x)^{1/2}}+C$ (4) None of these

9. $\int \frac{\sin x+\cos x}{9+16\sin 2x}dx$ is equal to

(1) $\frac{1}{20}\ln\left|\frac{5+4(\sin x-\cos x)}{5-4(\sin x-\cos x)}\right|+C$ (2) $\frac{1}{40}\ln\left|\frac{5+4(\sin x-\cos x)}{5-4(\sin x-\cos x)}\right|+C$

(3) $\frac{1}{20}\tan^{-1}\frac{4(\sin x-\cos x)}{5}+C$ (4) None to these

10. $\int e^x\frac{x-1}{(x+1)^3}dx$ is equal to

(1) $\frac{e^x}{(x+1)^2}+C$ (2) $\frac{e^x}{x+1}+C$

(3) $-\frac{e^x}{x+1}+C$ (4) $-\frac{e^x}{(x+1)^2}+C$

11. $\int \sqrt{\tan x}+\sqrt{\cot x}\,dx$

(1) $-\sqrt{2}\cot^{-1}\frac{\sqrt{\tan x}-\sqrt{\cot x}}{\sqrt{2}}+C$ (2) $\sqrt{2}\tan^{-1}\frac{\sqrt{\tan x}+\sqrt{\cot x}}{\sqrt{2}}+C$

(3) $\sqrt{2}\cot^{-1}\frac{\sqrt{\tan x}+\sqrt{\cot x}}{\sqrt{2}}+C$ (4) None of these

12. $\int \frac{dx}{x^{1/5}(1+x^{4/5})^{1/2}}$ is equal to

(1) $5(1+x^{4/5})^{1/2}+C$ (2) $\left(\frac{1}{2}\right)(1+x^{4/5})^{1/2}+C$

(3) $x^{4/5}(1+x^{4/5})^{1/2}+C$ (4) None of these

13. $\int \frac{\cos^4 x}{\sin^3 x(\sin^5 x+\cos^5 x)^{3/5}}dx =$

(1) $-\frac{1}{2}\left(\frac{\tan^5 x+1}{\tan^5 x}\right)^{2/5}+C$ (2) $-\frac{1}{5}\left(\frac{\tan^5 x+1}{\tan^5 x}\right)^{2/5}+C$

(3) $-\frac{1}{2}\left(\frac{\tan^{3/5} x+1}{\tan^5 x}\right)^{5/3}+C$ (4) None of these

14. $\int \frac{\cos 5x+\cos 4x}{1-2\cos 3x}dx = k\sin x(1+\cos x)+C,$ where $k$ is equal to

(1) 2 (2) –2

(3) 1 (4) –1

15. Let $\int x^{1/2}/(1-x^3)^{1/2}dx = (2/3)gof(x)+C$, then

(1) $f(x)=(x)^{1/2}$ (2) $f(x)=x^{3/2}$

(3) $f(x)=x^{2/3}$ (4) $g(x)=\sin^{-1}x$

16. $\int \frac{dx}{(1+\sqrt{x})^8} = \frac{-1}{3(1+\sqrt{x})^{k_1}}+\frac{2}{7(1+\sqrt{x})^{k_2}}+C,$ where

(1) $k_1=6$ (2) $k_2=7$

(3) $k_1=-6$ (4) $k_2=-7$

17. If $f(x)=\lim_{n\to\infty} e^{x\tan\left(\frac{1}{n}\right)\log\left(\frac{1}{n}\right)}$ and $\int \frac{f(x)}{\sqrt[3]{\sin^{11}x\cos x}}dx = g(x)+C,$ then

(1) $g\left(\frac{\pi}{4}\right)=\frac{15}{8}$

(2) $g(x)$ is continuous for all $x$

(3) $g\left(\frac{\pi}{4}\right)=-\frac{15}{8}$

(4) $g(x)$ is not differentiable at infinitely many point

Let $I_{n,m} = \int \sin^n x \cos^n \cdot dx$, the we can relate $I_{n,m}$ with each of the following

(i) $I_{n-2,m}$

(ii) $I_{n+2,m}$

(iii) $I_{n,m-2}$

(iv) $I_{n,m+2}$

(v) $I_{n-2,m+2}$

(vi) $I_{n+2,\ m-2}$

Suppose we want to establish a relation between $I_{n,m}$ and $I_{n,m-2}$, then we get

$P(x) = \sin^{n+1} x \cos^{m-1} x$ ...(i)

In $I_{n,m}$ and $I_{n,m-2}$ the exponent of cos$x$ is $m$ and $m-2$ respectively, the minimum of the two is $m-2$, adding 1 to the minimum we get $m-2+1 = m-1$. Now choose the exponent $m-1$ to cos$x$ in $P(x)$. Similarly choose the exponent of sin$x$ for $P(x)$. Now differentiating both sides of (i), we get

$$\begin{aligned} P'(x) &= (n+1)\sin^n x \cos^m x - (m-1)\sin^{n+2} x \cos^{m-2} x \\ &= (n+1)\sin^n x \cos^m x - (m-1)\sin^n x(1-\cos^2 x)\cos^{m-2} x \\ &= (n+1)\sin^n x \cos^m x - (m-1)\sin^n x \cos^{m-2} x + (m-1)\sin^n x \cos^m x \\ &= (n+m)\sin^n x \cos^m x - (m-1)\sin^n x \cos^{m-2} x. \end{aligned}$$

Now, integrating both sides we get $\sin^{n+1} x \cos^{m-1} x = (n+m)I_{n,m} - (m-1)I_{n,m-2}$.

18. The relation between $I_{4,\,2}$ and $I_{2,\,2}$ is

(1) $I_{4,2} = \frac{1}{6}(-\sin^3 x \cos^3 x + 3I_{2,2})$ (2) $I_{4,2} = \frac{1}{6}(\sin^3 x \cos^3 x + 3I_{2,2})$

(3) $I_{4,2} = \frac{1}{6}(\sin^3 x \cos^3 x - 3I_{2,2})$ (4) $I_{4,2} = \frac{1}{4}(-\sin^3 x \cos^3 x + 2I_{2,2})$

19. The relations between $I_{4,2}$ and $I_{6,2}$ is

(1) $I_{4,2} = \frac{1}{5}(\sin^5 x \cos^3 x + 8I_{6,2})$ (2) $I_{4,2} = \frac{1}{5}(-\sin^5 x \cos^3 x + 8I_{6,2})$

(3) $I_{4,2} = \frac{1}{5}(\sin^5 x \cos^3 x - 8I_{6,2})$ (4) $I_{4,2} = \frac{1}{6}(\sin^5 x \cos^3 x + 8I_{6,2})$

20. The relation between $I_{4,2}$ and $I_{4,4}$ is

(1) $I_{4,2} = \frac{1}{3}(\sin^5 x \cos^3 x + 8I_{44,})$ (2) $I_{4,2} = \frac{1}{3}(-\sin^5 x \cos^3 x + 8I_{44,})$

(3) $I_{4,2} = \frac{1}{3}(\sin^5 x \cos^3 x - 8I_{44,})$ (4) $I_{4,2} = \frac{1}{3}(\sin^5 x \cos^3 x + 6I_{44,})$

21. **Column I** **Column II**

(1) If $f(x)=\int\frac{x+\sin x}{1+\cos x}dx$ and $F(0)=0$, (a) $\frac{\pi}{2}$

then $f\left(\frac{\pi}{2}\right)=$

(2) Let $F(x)=\int e^{\sin^{-1}x}\left(1-\frac{x}{\sqrt{1-x^2}}\right)dx$ and (b) $\frac{\pi}{3}$

$F(0)=1$. If $F\left(\frac{1}{2}\right)=\frac{k\sqrt{3}e^{\pi/6}}{\pi}$, the $k$ equals

(3) Let $F(x)=\int\frac{dx}{(x^2+1)(x^2+9)}$ and $F(0)=0$, (c) $\frac{\pi}{4}$

If $F\left(\sqrt{3}\right)=\frac{5}{36}k$, then $k$ equals

(4) Let $F(x)=\int\frac{\sqrt{\tan x}}{\sin x\cos x}dx$ and $F(0)=0$; (d) $\pi$

if $F\left(\frac{\pi}{4}\right)=\frac{2k}{\pi}$, then $k$ equals

22. If $\int\frac{\sin x(\cos x)^{-5/2}}{\sqrt{\sin x+3\cos x}+\sqrt{\sin x+4\cos x}}dx=\frac{\lambda}{740}\left((\tan x+4)^{5/2}-(\tan x+3)^{3/2}\right)$

$-\frac{\mu}{3}\left[4(\tan x+4)^{3/2}-3(\tan x+3)^{3/2}\right]+C$, then evaluate $\frac{\lambda}{\mu}$.

23. If $\int\frac{x^3}{x^{16}+4}dx=\frac{1}{\lambda}\tan^{-1}\frac{z}{\sqrt{2}}-\frac{1}{\mu}\ln\left|\frac{u-\sqrt{2}}{u-\sqrt{2}}\right|+C$ where $u=y+\frac{1}{y}$ and $z=y-\frac{1}{y}, y=\frac{x^4}{\sqrt{2}}$, the evaluate $\lambda+\mu$.

24. $\int_{\pi/3}^{\pi/2}\frac{\sqrt{1+\cos x}}{(1-\cos x)^{5/2}}dx$ is equal to

(1) $\frac{5}{2}$ (2) $\frac{3}{2}$

(3) $\frac{1}{2}$ (4) $\frac{2}{5}$

25. $\int_0^{\pi/4}\frac{dx}{\cos^4 x-\cos^2 x\sin^2 x+\sin^4 x}$ is equal to

(1) $\frac{\pi}{2}$ (2) $\frac{\pi}{4}$

(3) $\frac{\pi}{3}$ (4) None of these

26. If $\int_0^{\pi} x f(\cos^2 x + \tan^4 x)dx = k\int_0^{\pi/2} f(\cos^2 x + \tan^4 x)dx$, then the value of $k$ is

(1) $\frac{\pi}{2}$ (2) $\pi$

(3) $-\frac{\pi}{2}$ (4) None of these

27. If $f(x)$ and $g(x)$ are continuous functions, then $\int_{\ln\lambda}^{\ln 1/\lambda} \frac{f\left(\frac{x^2}{4}\right)[f(x)-f(-x)]}{g\left(\frac{x^2}{4}\right)[g(x)+g(-x)]}dx$ is

(1) Dependent of $\lambda$ (2) A non-zero constant

(3) Zero (4) None of these

28. Let $\int_0^1 f(x)dx = 1, \int_0^1 x f(x)dx = a$ and $\int_0^1 x^2 f(x)dx = a^2$, then the value of $\int_0^1 (x-a)^2 f(x)dx =$

(1) 0 (2) $a^2$

(3) $a^2 - 1$ (4) $a^2 - 2a + 2$

29. If $\int_0^1 \frac{e^t dt}{t+1} = a$, then $\int_{b-1}^{b} \frac{e^{-t}dt}{t+b-1}$ is equal to

(1) $ae^{-b}$ (2) $-be^{-a}$

(3) $-ae^{-b}$ (4) None of these

30. If $f : \mathbb{R} \to \mathbb{R} : f(x) = \sin x + x$, then the value of $\int_0^{\pi} (f^{-1}(x))dx$, is equal to

(1) $\frac{\pi^2}{2} - 2$ (2) $\pi^2$

(3) $\frac{\pi^2}{2} + 2$ (4) None of these

31. $\int_{-1/2}^{1/2} \sqrt{\left\{\left(\frac{x+1}{x-1}\right)^2 + \left(\frac{x-1}{x+1}\right)^2 - 2\right\}}dx$ is

(1) $4\log\frac{3}{4}$ (2) $4\log\frac{4}{3}$

(3) $2\log\frac{16}{9}$ (4) None of these

32. If $I=\int_0^1 \cos\left(2\cot^{-1}\sqrt{\left(\frac{1-x}{1+x}\right)}\right)dx$, then

(1) $I>\frac{1}{2}$ (2) $I=-\frac{1}{2}$

(3) $0<1<\frac{1}{2}$ (4) None of these

33. The value of $\int_{-\sqrt{2}}^{\sqrt{2}} \frac{2x^7+3x^6-10x^5-7x^3-12x^2+x+1}{x^2+2}dx$ is

(1) $2\int_0^{\sqrt{2}} \frac{3x^6-12x^2+1}{x^2+2}dx$ (2) 0

(3) $\frac{-14}{5}\sqrt{2}+\frac{\pi}{2\sqrt{2}}$ (4) $\frac{-16}{5}\sqrt{2}-\frac{\pi}{2\sqrt{2}}$

34. The value of the integral $\int_0^{\infty} \frac{x\log x}{(1+x^2)^2}dx$ is

(1) 0 (2) 7

(3) 5log13 (4) 2log5

35. The set of all the values of $\alpha$ for which the inequality $\frac{1}{\sqrt{\alpha}}\int_1^{\alpha}\left(\frac{3}{2}\sqrt{x}+1-\frac{1}{\sqrt{x}}\right)dx<4$ holds is given by

(1) [0, 4] (2) (0, 4)

(3) (0, 1) (4) None of these

36. The value of the integral $\int_0^{\pi} \frac{x^2\sin x}{(2x-\pi)(1+\cos^2 x)}dx$ is

(1) $\frac{\pi^2}{4}$ (2) $\frac{\pi^2}{2}$

(3) $\frac{\pi^2}{6}$ (4) None of these

37. If $f:\mathbb{R}\to\mathbb{R}$ is continuous and differentiable function such that $\int_{-1}^{x} f(t)dt+f''(3)\int_x^0 dt=\int_1^x t^3dt-f'(1)\int_0^x t^2dt+f''(2)\int_x^3 tdt$, then the value of $f'(4)$ is

(1) $48-8f'(1)-f''(2)$ (2) $48+8f'(1)-f''(2)$

(3) $48-8f'(1)+f''(2)$ (4) $48+8f'(1)+f''(2)$

38. Let $I_1 = \int_{\sec^2 z}^{2-\tan^2 z} xf(x(3-x))dx$ and $I_2 = \int_{\sec^2 z}^{2-\tan^2 z} f(x(3-x))dx$; where $f$ is a continuous function and $z$ is any real number, then $\frac{I_1}{I_2} =$

(1) $\frac{3}{2}$ (2) $\frac{1}{2}$

(3) 1 (4) None of these

39. If $I = \int_1^2 \frac{dx}{\sqrt{2x^3 - 9x^2 + 12x + 4}}dx$, then

(1) $\frac{1}{2} < I < \frac{1}{3}$ (2) $\frac{1}{4} < I < \frac{1}{3}$

(3) $\frac{1}{4} < I < 1$ (4) None of these

40. If $\int_0^1 \frac{\sin t}{1+t}dt = \alpha$, then the value of the integral $\int_{4\pi-2}^{4\pi} \frac{\sin t/2}{4\pi+2-t}dt$ in terms of $\alpha$ is given by

(1) $2\alpha$ (2) $-2\alpha$

(3) $\alpha$ (4) $-\alpha$

41. $\int_0^1 \frac{dx}{1+x^2+2x^5}$ lies between

(1) $\frac{\pi}{6\sqrt{3}}$ and $\frac{\pi}{4}$ (2) $\frac{\pi}{3\sqrt{3}}$ and $\frac{\pi}{2}$

(3) $\frac{\pi}{3\sqrt{3}}$ and $-\frac{\pi}{4}$ (4) None of these

42. If $A = \int_1^{\sin\theta} \frac{tdt}{1+t^2}$ and $B = \int_1^{\text{cosec}\theta} \frac{dt}{t(1+t^2)}$, then the value of $\begin{vmatrix} A & A^2 & B \\ e^A e^B & B^2 & -1 \\ 1 & A^2+B^2 & -1 \end{vmatrix}$ is

(1) $\sin\theta$ (2) $\text{cosec}\theta$

(3) 0 (4) 1

43. If $\int_0^{\pi/2} \frac{dx}{a^2\cos^2 x + b^2\sin^2 x} = \frac{\pi}{2ab}$, then the value of $\int_0^{\pi/2} \frac{dx}{(4\cos^2 x + 9\sin^2 x)^2}$

(1) $\frac{11\pi}{864}$ (2) $\frac{13\pi}{864}$

(3) $\frac{17\pi}{864}$ (4) None of these

44. State which of the following statements are true?

(1) If $m$ and $M$ are the least and the greatest value of $f(x)$ in $[a, b]$, then $m(b-a) \le \int_a^b f(x)dx \le M(b-a)$

(2) If $f^2(2)$ and $g^2(x)$ are integrable on $[a, b]$, then $\left|\int_a^b f(x)g(x)dx\right| \le \left(\int_a^b f^2(x)dx\right)^{1/2}\left(\int_a^b g^2(x)dx\right)^{1/2}$

(3) If $f(x)$ is continuous on $[a, b]$, then there exists a point $c \in (a, b)$, such that $\int_a^b f(x)dx = f(c)(b-a)$

(4) If $f(x)$ is periodic function with period $T$ and $a \in R^+$, then $\int_a^{a+T} f(x)dx = \int_0^T f(x)dx$

45. If $I_n = \int_0^{\pi/4} \tan^n x dx, n \in W$

(1) $I_n + I_{n-2} = \frac{1}{n-1}$ for $n \ge 2$

(2) $I_{n+1} + I_{n-2} = \frac{1}{n-1}$ for $n \ge 2$

(3) $\frac{1}{n+1} < 2I_n < \frac{1}{n-1}$; where $n > 1$

(4) $(I_{n+2} + I_n)(n+1) = 1$ for $n \ge 1$

46. Let $I_n = \int_0^{\pi/2} \cos^n x\, dx, n \in W$. Then

(1) $I_{n-2} > I_n \ \forall\ n \ge 2$

(2) $n(I_{n-2} - I_n) = I_{n-2} \ \forall\ n \ge 2$

(3) $I_n : I_{n-1} = n : (n-1) \ \forall\ n \ge 1$

(4) None of these

47. If $I_n = \int_0^1 \frac{dx}{(1+x^2)^n}, n \in \mathbb{N}$, then which of the following statements hold good?

(1) $2nI_{n+1} = 2^{-n} + (2n-1)I_{n-1}$

(2) $I_2 = \frac{\pi}{8} + \frac{1}{4}$

(3) $I_2 = \frac{\pi}{8} - \frac{1}{4}$

(4) $I_3 = \frac{\pi}{16} - \frac{5}{48}$

48. If $f(2-x) = f(2+x)$ and $f(4-x) = f(4+x)$ and $f(x)$ is a function for which $\int_0^2 f(x)dx = 5$, then $\int_0^{50} f(x)dx$ is equal to

(1) 125

(2) $\int_{-4}^{46} f(x)dx$

(3) $\int_1^{51} f(x)dx$

(4) $\int_2^{52} f(x)dx$

49. If $f(x)=\int_0^x 2|t|dt$, the tangent lines which are parallel to the bisector of the first quadrant angle is

(1) $y=x-\frac{1}{4}$ (2) $y=x-\frac{3}{2}$

(3) $y=x+\frac{3}{2}$ (4) $y=x+\frac{1}{4}$

50. Let $f(x)$ be a non constant twice differentiable function defined on $(-\infty,\infty)$ such that $f(x) = f(1 - x)$ and $f'\left(\frac{1}{4}\right)=0$. Then

(1) $f'(x)$ vasnishes at least twice on [0, 1] (2) $f'\left(\frac{1}{2}\right)=0$

(3) $\int_{-1/2}^{1/2} f\left(x+\frac{1}{2}\right)\sin x\,dx=0$ (4) $\int_0^{1/2} f(t)e^{\sin \pi t}dt=\int_{1/2}^{1} f(1-t)e^{\sin \pi t}dt$

51. The value of $\int_{-\pi}^{\pi} \sin mx\cos nx dx(m,n\in\mathbb{N})$ is equal to

(1) 0 (2) $\int_{-\pi/8}^{\pi/8} x^8\sin^9 x dx$

(3) $\int_{-\pi/2}^{\pi/2} \sin x f(\cos x)dx$ (4) None of these

52. The value of $\int_3^4 \frac{dx}{\sqrt[3]{\log x}}$ is

(1) Less than one (2) Greater than 1/2

(3) Less than two (4) Greater than 1/4

If $f(x, a)$ be a continuous function of $x$ and $\alpha\forall x\in[a,b]$ and $\alpha\in[c,d]$ where $x$ and $\alpha$ be two independent parameters and $I(\alpha)=\int_a^b f(x,\alpha)dx$ the $I'(\alpha)=\frac{dI(\alpha)}{d\alpha}=\int_a^b\left(\frac{\partial}{\partial\alpha}f(x,\alpha)\right)dx$. That is, integrating $f(x, \alpha)$ with respect to $x$ and then differentiating the result wit respect to $\alpha$ results into same output as when we differentiate $f(x, \alpha)$ with respect to $\alpha$ first and then integrate the result with respect to parameter $x$. e.g. $f(x, \alpha) = 3\alpha^2x^2 + 2\alpha x + \alpha$

$$I(\alpha)=\int_0^1 f(x,\alpha)dx=[\alpha^2x^3+\alpha x^2+\alpha x+c]_0^1$$
$$=\alpha^2+\alpha+\alpha\Rightarrow I'(\alpha)=2\alpha+2$$

$$f'(x,\alpha)=\frac{d}{d\alpha}f(x,\alpha)=6\alpha x^2+2x+1$$

$$\Rightarrow \int_0^1 f'(x,\alpha)dx=\left[\frac{6\alpha x^3}{3}+\frac{2x^2}{2}+x\right]_0^1=2\alpha+2$$

Based on the above theorem answer the following problems:

53. Let $I(\alpha,k)=\int_0^\infty \frac{e^{-\alpha x}\sin kx}{3x}dx$, then $\int_0^\infty \frac{\sin 2x}{3x}dx$ is equal to; $\alpha > 0$

(1) $\frac{\pi}{2}$ (2) Not integrable

(3) $\frac{\pi}{6}$ (4) None of these

54. If the value of integral $\int_0^\pi \frac{dx}{a-\cos x}=\frac{\pi}{2\sqrt{a^2-1}}$, the value of integral $\int_0^\pi \frac{dx}{(a-\cos x)^2}$ is equal to

(1) $\frac{\pi a}{2(a^2-1)^{1/2}}$ (2) $\frac{\pi a}{2(a^2-1)^{3/2}}$

(3) $\frac{\pi a}{(a^2-1)^{3/2}}$ (4) None of these

55. The value of the integral = $\int_0^1 \frac{\ln(1-a^2x^2)}{x^2\sqrt{(1-x^2)}}dx$ ($a^2 < 1$) is equal to

(1) $\frac{\pi}{\sqrt{1-a^2}-1}$ (2) $\frac{\pi}{\sqrt{1-a^2}+1}$

(3) $\pi\left(\sqrt{1-a^2}-1\right)$ (4) $\pi\left(\sqrt{1-a^2}+1\right)$

Let [*a*, *b*] be an interval that does not contain 0 in its interior and let *f* be a non-negative continuous function on [*a*, *b*] (say $a > 0$). If the region bounded by the graph of *f* and the *x*-axis is revolved around *y*-axis, then a solid is generated. The volume of this solid is given $V=2\pi\int_a^b xf(x)dx$. On the basis of above information answer the following questions:

56. The region bounded by $f(x) = 4x - x^2$ and the *x*-axis between $x = 1$ and $x = 4$ is rotated about the *y*-axis. The volume of the solid generated is

(1) $\frac{81}{2}\pi$ (2) $\frac{41}{2}\pi$

(3) $\frac{83}{2}\pi$ (4) None of these

57. Let $f(x)=\begin{cases}\sqrt{3}x; & 0\le x<1\\ \sqrt{4-x^2}; & 1\le x\le 2\end{cases}$ and $A$ be the region between the graph of $f$ and $x$-axis. On revolving $A$ around $y$-axis, the volume of the solid, thus obtained is

(1) $\frac{4\sqrt{3}}{3}\pi$ (2) $\frac{8\sqrt{3}}{3}\pi$

(3) $\frac{2\sqrt{3}}{3}\pi$ (4) None of these

58. An ellipse whose major axis is $2a$ and the mirror axis is $2b$ revolves about the minor axis. The volume of the solid is equal to

(1) $\frac{4}{3}\pi ab^2$ (2) $\frac{2}{3}\pi ab^2$

(3) $\frac{4}{3}\pi a^2 b$ (4) None of these

59. If $F(x)=\int_0^x f(t)dt$, then

| **Column I** | **Column II** |
|---|---|
| (1) $f(x)=\begin{cases}\cos\frac{\pi}{2}x; & 0\le x<1\\ \sin\pi x; & x\ge 1\end{cases}$ | (a) $F'(2)=1$ |
| (2) $f(x)=\begin{cases}\sqrt{3-x}; & 0\le x\le 2\\ (5x-9)^{1/3} & x>2\end{cases}$ | (b) $F'(7.2)=3$ |
| (3) $f(x)=\begin{cases}x^4; & 0\le x\le 1\\ \cos 2\pi x; & x>1\end{cases}$ | (c) $F'\left(\frac{3}{2}\right)=-1$ |
| (4) $f(x)=\begin{cases}\sin\frac{3\pi}{2}x; & 0\le x<1\\ -\sqrt{3-2x}; & \frac{3}{2}\ge x>1\end{cases}$ | (d) $F'\left(\frac{3}{2}\right)=-0$ |

60. If $\int_0^{\pi/2}\frac{dx}{\left(\sqrt{\cos x}+\sqrt{\sin x}\right)^4}=\lambda$, then evaluate $\lambda^{-1}$.

61. If $\int_0^{\pi/4}\frac{\ln(\cot x)\cdot(\sin 2x)^{2008}dx}{[(\sin x)^{2009}+(\cos x)^{2009}]^2}\frac{(\lambda)^{\mu}\ln\lambda}{\nu^2}$, then evaluate $(\lambda+\mu+\nu)$

62. Evaluate $\lim_{n\to\infty}\frac{(m+1)}{n^{(m+1)}}(1^m+2^m+3^m+...+n^m);\ m>-1$

63. Evaluate $\lim_{n\to\infty}\left[C_0^2 + C_1^2 + C_2^2 + ... + C_n^2\right]^{1/n}$; where $C_r = {}^nC_r$

64. Find $\lim_{x\to 0}\frac{\int_{-x}^{x} f(t)dt}{\int_{0}^{2x} f(t+4)dt}$; where $f$ is a contiunous function and $f(x) > 0$ for all $x$ satisfying $f(0) = 48f(4)$.

65. If $I = 636\int_{0}^{\pi/4}\left(\sqrt{\sin x} + \sqrt{\cos x}\right)^{-4} dx$, then find $I$.

66. Area bounded by the curve $y = xe^{x^2}$, $x$-axis and the ordinates x = 0, $x = a$ is

(1) $\frac{e^{a^2}-1}{2}$ Square unit (2) $\frac{e^{a^2}+1}{2}$ Square unit

(3) $\left(e^{a^2}+1\right)$ Square unit (4) $\left(e^{a^2}-1\right)$ Square unit

67. The area bounded by the $x$-axis, the curve $y = f(x)$ and the lines $x = 1$, $x = b$ is equal to $\sqrt{b^2+1} - \sqrt{2}$ for all $b > 1$, then $f(x)$ is

(1) $\sqrt{x-1}$ (2) $\sqrt{x+1}$

(3) $\sqrt{x^2+1}$ (4) $\frac{x}{\sqrt{1+x^2}}$

68. If the ordinate $x = a$ divides the area bounded by the curve $y = 1 + \frac{8}{x^2}$ and the ordinates $x = 2$, $x = 4$ into two equal parts, then $a$ is equal to

(1) $2\sqrt{3}$ (2) $2\sqrt{2}$

(3) 3 (4) None of these

69. If $f(x) = \begin{cases} \sqrt{\{x\}}; & x \notin \mathbb{Z} \\ 1; & x \in \mathbb{R} \end{cases}$ and $g(x) = (x)^2$, (where $\{\cdot\}$ denotes fractional part of $x$), then area bounded by $f(x)$ and $g(x)$ for $x \in [0, 10]$ is

(1) $\frac{5}{3}$ (2) 5

(3) $\frac{10}{3}$ (4) None of these

70. Let $f(x) = x^2$, $g(x) = \cos x$ and $\alpha$, $\beta$ ($\alpha < \beta$) be the roots of the equation $18x^2 - 9\pi x + \pi^2 = 0$. Then the area bounded by the curves $y = fog(x)$, the ordinates $x = \alpha$, $x = \beta$ and the $x$-axis is

(1) $\frac{1}{2}(\pi - 3)$ (2) $\frac{\pi}{3}$

(3) $\frac{\pi}{4}$ (4) $\frac{\pi}{12}$

71. Area included between the parabolas $y = \frac{x^2}{4a}; a > 0$ and the curve $y = \frac{8a^3}{x^2 + 4a^2}; a > 0$ is

(1) $\frac{a^2}{6}(6\pi - 4)$ (2) $\frac{a^2}{3}(4\pi + 3)$

(3) $\frac{a^2}{3}(8\pi + 3)$ (4) None of these

72. The area bounded by $x^2 + y^2 - 2x = 0$ and $y = \sin\frac{\pi x}{2}$ in the upper half of the circle is

(1) $\frac{\pi}{2} - \frac{4}{\pi}$ (2) $\frac{\pi}{4} - \frac{2}{\pi}$

(3) $\pi - \frac{8}{\pi}$ (4) None of these

73. The area bounded by the curves $y = x(1 - \ln x)x = e^{-1}$ and a positive $x$-axis between $x = e^{-1}$ and $x = e$ is

(1) $\left(\frac{e^2 - 4e^{-2}}{5}\right)$ (2) $\left(\frac{e^2 - 5e^{-2}}{4}\right)$

(3) $\left(\frac{4e^2 - e^{-2}}{5}\right)$ (4) $\left(\frac{5e^2 - e^{-2}}{4}\right)$

74. The area of the figure bounded by the curve $(y - x - 2)^2 = 9x$, $y$-axis and $x$-axis is

(1) $\frac{1}{2}$ (2) $\frac{3}{4}$

(3) 1 (4) None of these

75. A figure is bounded by the curves $y = x^2 + 6x + 9$ and the co-ordinate axes. The angles which the straight lines passing through the point (0, 9) must make with positive semi axes so that they partition the figure into three parts of equal areas are

(1) $\cos^{-1}\frac{27}{2}$ and $\cos^{-1}\frac{27}{4}$ (2) $\sin^{-1}\frac{27}{2}$ and $\sin^{-1}\frac{27}{4}$

(3) $\tan^{-1}\frac{27}{2}$ and $\tan^{-1}\frac{27}{4}$ (4) None of these

76. Consider the graph of continuous function $y = f(x)$ for $x \in [a - b, a + b]$ $a, b \in R^+$ and $b > a$. If the origin is shifted to $(a + b, 0)$ such that new axes are parallel to the old axes, then the area bounded by the given curve, the $X$-axis and the new ordinates $X = -a$, $X = -b$ can be written as

(1) $\int_{\frac{b-a}{2}}^{\frac{a+b}{2}} f(x)dx$ (2) $\int_{-a}^{-b} f(x)dx$

(3) $\int_{a}^{b} f(a + b - x)dx$ (4) $\frac{1}{2}\int_{a-b}^{a+b} f(x)dx$

77. $P$ is any point inside or on the boundary of $\Delta ABC$ having perimeter $p$ and area $\Delta$. $R$ is any point in the plane of $\Delta ABC$ such that $PR \le 5$. The area of the region in which the point $R$ lies is

(1) $25\pi + 5p + \Delta$ (2) $5\pi + 25p + \Delta$

(3) $5\pi + 5p + \Delta$ (4) $5(\pi + p) + 25\Delta$

78. The are bounded by $y = f(x)$, $x$-axis and the line $y = 1$, where $f(x) = 1 + \frac{1}{x}\int_1^x f(t)\,dt$ is

(1) $2(e + 1)$ (2) $2\left(1 - \frac{1}{e}\right)$

(3) $2(e - 1)$ (4) None of these

79. The area bounded by the curves $y = \frac{\sin x}{e^x} \forall x \in [0, 2\pi]$ and the axes of abscissae

(1) $\frac{e^{-2\pi}}{2} - e^{-\pi} + \frac{1}{2}$ (2) $\frac{e^{-2\pi}}{2} + e^{-\pi} - \frac{1}{2}$

(3) $\frac{e^{-2\pi}}{2} + e^{-\pi} + \frac{1}{2}$ (4) None of these

80. The area bounded by the curves $y = \sin^{-1}|\sin x|$ and $y = (\sin^{-1}|\sin x|)^2$, $0 \le x \le 2\pi$ is

(1) $\left(\frac{\pi^3}{3} + \frac{4}{3}\right)$ square unit (2) $\left(\frac{\pi^3}{6} - \frac{\pi^2}{2} + \frac{4}{3}\right)$ square unit

(3) $\left(\frac{\pi^2}{2} - \frac{4}{3}\right)$ square unit (4) $\left(\frac{\pi^3}{6} - \frac{\pi^2}{4} + \frac{4}{3}\right)$ square unit

81. The area contained between the $x$-axis and one arc of the curve $y = \cos 9x$ is:

(1) $\frac{1}{3}$ (2) $\frac{2}{3}$

(3) $\frac{2}{7}$ (4) $\frac{2}{9}$

82. Area of the loop of the curve $x = a(1 - t^2)$, $y = at(1 - t^2)$, $-1 \le t \le 1$ must be

(1) $\frac{2}{15}a^2$ (2) $\frac{4}{15}a^2$

(3) $\frac{8}{15}a^2$ (4) $\frac{a^2}{5}$

83. The area bounded between the curves $x^2 = y$ and $y^2 = x$ is equal to

(1) Area of the region $\{(x, y) : x^2 \le y \le |x|\}$ (2) $\frac{1}{3}$

(3) $2\int_0^1 (x - x^2)\,dx$ (4) None of these

84. The area of quadrilateral, combined equation of whose sides are $(x^2 - y^2)(x^2 - 8x + 6) = 0$, is

(1) 8 (2) >5

(3) $2\sqrt{2}$ (4) 9

85. If $A\left(\frac{3}{\sqrt{2}}, \sqrt{2}\right), B\left(\frac{-3}{\sqrt{2}}, \sqrt{2}\right), C\left(\frac{-3}{\sqrt{2}}, -\sqrt{2}\right)$ and $D(3\cos\theta, 2\sin\theta)$ are four points, then the value of $\theta$ for which the area of quadrilateral *ABCD* is maximum, $\left(\frac{3\pi}{2} \le \theta \le 2\pi\right)$ is

(1) $2\pi - \sin^{-1}\frac{1}{3}$ (2) $\frac{7\pi}{4}$

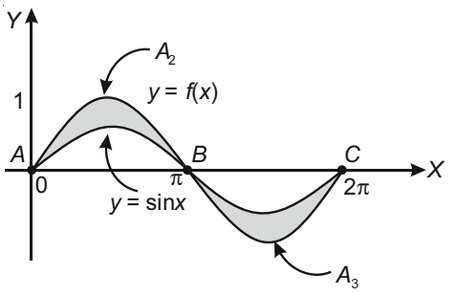
(3) $2\pi - \cos^{-1}\frac{3}{\sqrt{85}}$ (4) $\tan\left(\tan^{-1}\frac{17\pi}{4}\right)$

86. Let $f$ and $g$ be continuous function on $a \le x \le b$ and set $p(x) = \max\{f(x), g(x)\}$ and $q(x) = \min\{f(x), g(x)\}$, the area bounded by the curves $y = p(x)$, $y = q(x)$ and the ordinates $x = a$ and $x = b$ is given by

(1) $\int_a^b (f(x) - g(x))dx$ (2) $\int_a^b (p(x) - q(x))dx$

(3) $\int_a^b |(p(x) - q(x))|dx$ (4) $\int_a^b |(f(x) - g(x))|dx$

In the adjacent figure, the graphs of two functions $y = f(x)$ and $y = \sin x$ are given. They intersect at origin, $A(a, f(\alpha))$, $B(\pi, 0)$ and $C(2\pi, 0)$. $A_i (i = 1, 2, 3)$ is the area bounded by the curves as shown in the figure respectively for $x \in (0, \alpha)$, $x \in (\alpha, \pi)$, $x \in (\pi, 2\pi)$. If $A_1 = 1 + (\alpha - 1)\cos\alpha - \sin\alpha$. On the basis of above information, answer the following questions.




87. The functinos $f(x)$ is

(1) $x^2\sin x$ (2) $x\sin x$

(3) $2x\sin x$ (4) $x^3\sin x$

88. Value of $A_2$ for $a = 1$, is

(1) $(\pi - 1)$ units$^2$ (2) $\left(\frac{\pi}{2} - 1\right)$ units$^2$

(3) $(\pi - \sin 1 - 1)$ (4) $\frac{\pi}{2}$ units$^2$

89. Value of $A_3$ is

(1) $(2\pi - 1)$ units$^2$ (2) $(3\pi - \sin 2)$ units$^2$

(3) $(3\pi - 2)$ units$^2$ (4) $(\pi - 2)$ units$^2$

Let $A$, $O$, $B$ and $C$ are fixed points on a straight line such that $AO = OB = BC = 1$ unit. The points $A$ and $B$ are also joined by a semicircle of radius 1 unit, and $P$ is a variable point on this semicircle such that $\angle POC = \theta$. Consider the region $R$ bounded by the arc $AP$ of the semi circle and the straight lines $PC$ and PC.

90. Area of region $R$ is maximum when $\theta$ is

(1) $\frac{\pi}{6}$ (2) $\frac{\pi}{2}$

(3) $\frac{\pi}{3}$ (4) $\pi$

91. Minimum value of perimeter of region $R$ is

(1) 3 units (2) $\pi$ units

(3) 4 units (4) 6 units

92. The graphs of area and perimeter of the region $R$ are being drawn, w.r.t., $\theta$. Tangents to the two graphs are parallel

(1) No where (2) At least once

(3) At least twice (4) At least thrice

93.

| **Column I** | **Column II** |
|---|---|
| (1) Let area of the figure bounded by $y = -3x^2 - \lvert x \rvert + 2$ and $y = 0$ is $\frac{22}{a^3}$ square units then $a$ is | (a) 4 |
| (2) Let $I_{n,m} = \int \frac{\sin^n x}{\cos^m x} dx$ $m \neq 1$, If $I_{n,m} = \frac{\sin^{n-1} x}{(m-1)\cos^{m-1} x} + \frac{a(n-1)}{2(m-1)} I_{n-2,m-2}$ then $a$ is | (b) 3 |
| (3) Let $f(x) = x - x^2$; $g(x) = ax$; If the area of the region above the graph of $g(x)$ and below the graph of $f(x)$ is $\frac{9}{2}$ square units, the $a$ is | (c) –2 |
| (4) Let $f_n(x) = \int \cot^n x \, dx$, then $3\left(f_2\left(\frac{3\pi}{4}\right) + f_4\left(\frac{3f}{4}\right)\right)$ is | (d) 1 |

94. The area bounded by the cruve $y = \sqrt{x - [x]}$, between the ordinates $x = 1$, $x = \frac{3}{2}$ and $x$-axis is (here [ ] is the greatest integer function) is $\frac{1}{3}\sqrt{k}$, then evaluate $k$.

95. Area bounded by the curve $y = \sqrt{\sin[x] + [\sin x]}$; (where [ ] is greatest integer function), lines $x = 1$ and $x = \frac{\pi}{2}$ and the $x$-axis is given by $\sqrt{\sin k}\left(\frac{\pi}{2} - k\right)$, then evaluate $k$.

96. Let $f(x)$ be a twic differentiable function defined as $R \to R$ such that the equatin $\lambda^2 f(x) - 2\lambda f'(x) + f''(x) = 0$ provides two equal values of $\lambda$ for all $x$. If $f(0) = 1$ and $f'(0) = 2$, then find the area bounded by the curve $y = f(x)$ and $x$-axis in the region $x + y < 1$.

97. Let '$c$' be the constant number such that $c > 1$. If the least ara of the figure given by the line passing through the point $(1, c)$ with gradient '$m$' and the parabola $y = x^2$ is 36 sq. units, find the value of $(c^2 + m^2)$.

98. If the area bounded by the curves $y = \cos^{-1}x$; $y = \sin^{-1}x$ and $y = -\pi x^3$; where $-1 \le x \le 1$, is $A$ sq. units, then find $[A]$. Where [A] represents integra value of $A$.

99. The degree of the differential equation $y(x) = 1 + \frac{dy}{dx} + \frac{1}{1.2}\left(\frac{dy}{dx}\right)^2 + \frac{1}{1.2.3}\left(\frac{dy}{dx}\right)^3 + \ldots$ is

(1) 2 (2) 3

(3) 1 (4) None of these

100. If $m$ and $n$ are order and degree of the equation $\left(\frac{d^2y}{dx^2}\right)^5 + 4 \cdot \frac{\left(\frac{d^2y}{dx^2}\right)}{\frac{d^3y}{dx^3}} + \frac{d^3y}{dx^3} = x^2 - 1$, then

(1) $m = 3$, $n = 3$ (2) $m = 3$, $n = 2$

(3) $m = 3$, $n = 5$ (4) $m = 3$, $n = 1$

101. The differential equation of all conics whose centre lies at origin, is given by

(1) $(3xy_2 + x^2y_3)(y - xy_1) = 3xy_2(y - xy_1 - x^2y_2)$ (2) $(3xy_1 + x^2y_2)(y_1 - xy_3) = 3xy_1(y - xy_2 - x^2y_3)$

(3) $(3xy_2 + x^2y_3)(y_1 - xy) = 3xy_1(y - xy_1 - x^2y_2)$ (4) None of these

102. The differential equatin of circles passing through the points of intersection of unit circle with centre at the origin and the line bisecting the first quadrant, is

(1) $y_1(x^2 + y^2 - 1) + (x + yy_1) = 0$ (2) $(y_1 - 1)(x^2 + y^2 - 1) + 2(y - x)(x + yy') = 0$

(3) $(x^2 + y^2 - 1) + yy_2 = 0$ (4) None of these

103. The solution of $\frac{dy}{dx} = (x + y - 1) + \frac{x + y}{\log(x + y)}$ is given by

(1) $\{1 + \log(x + y)\} - \log\{1 + \log(x + y)\} = x + c$

(2) $\{1 + \log(x + y)\} - \log\{1 - \log(x + y)\} = x + c$

(3) $\{1 + \log(x + y)\}^2 - \log\{1 + \log(x + y)\} = x + c$

(4) None of these

104. The solution of $ydx - xdy + (1 + x^2)dx + x^2 \sin ydy = 0$, is given by:

(1) $x + 1 - y^2 + \cos y + c = 0$ (2) $y + 1 - x^2 + x\cos y + c = 0$

(3) $\frac{x}{y} + \frac{1}{y} - y + \cos y + c = 0$ (4) $\frac{y}{x} + \frac{1}{x} - x + \cos y + c = 0$

105. The solution of $\left(\frac{dy}{dx}\right)^2 + (2x + y) - \frac{dy}{dx} + 2xy = 0,$ is

(1) $(y + x^3 - c_1)(x + \log y + y^2 + c_2) = 0$ (2) $(y + x^3 - c_1)(x - \log y - c_2) = 0$

(3) $(y + x^3 - c_1)(x - \log y - c) = 0$ (4) None of these

106. The degree of the differential equation $\frac{d^2y}{dx^2} + 3\left(\frac{dy}{dx}\right)^2 = x\log\frac{d^2y}{dx^2}$ is

(1) 1 (2) 2

(3) 3 (4) None of these

107. The orthogonal trajectory of the curve $a^{n-1}y = x^n$ is

(1) $nx^2 + y^2 = 0$ (2) $ny + x^2 =$ constant

(3) $y^2 - x^2 =$ constant (4) None of these

108. Which of the following transformationreduce the differential equation $\frac{dz}{dx} + \frac{z}{x}\log z = \frac{z}{x^2}(\log z)^2$ into the form $\frac{du}{dx} + P(x)u = Q(x)$?

(1) $u = \log z$ (2) $u = e^z$

(3) $u = (\log z)^{-1}$ (4) $u = (\log z)^2$

109. If each $a_i > 0$, then the shortest distance between the point (0, –3) and the curve $y = 1 + a_1x^2 + a_2x^4 + ... + a_nx^{2n}$ is

(1) 1 (2) 2

(3) 3 (4) 4

110. Solution of differential equation $\frac{x^2}{\left(\frac{1}{x^2} + y^2\right)}(xdy + ydx) + y^2(xdy - ydx) = 0$ is

(1) $\tan^{-1}xy - \frac{1}{2}\left(\frac{y}{x}\right)^2 = c$ (2) $\tan^{-1}xy + \frac{1}{2}\left(\frac{y}{x}\right)^2 = c$

(3) $\tan^{-1}xy + \frac{1}{2}\left(\frac{y}{x}\right)^3 = c$ (4) None of these

111. The differential equatin satisfying the curve $\frac{x^2}{a^2 + \lambda} + \frac{y^2}{b^2 + \lambda} = 1,$ is

(1) $(x + yy_1)(xy_1 - y) = (a^2 - b^2)y_1$ (2) $(x + yy_1)(xy_1 - y) = y_1$

(3) $(x - yy_1)(xy_1 + y) = (a^2 - b^2)y_1$ (4) None of these

112. If $y_1$, $y_2$ are the solutions of the differential equation $\frac{dy}{dx} + p(x)y = Q(x)$, then

(1) $y = y_1 + C(y_1 - y_2)$ is the general solution of equation

(2) $y = y_1 + C(y_1 + y_2)$ is the general solution of equation

(3) $\alpha y_1 + \beta y_2$ is a solution if $\alpha + \beta = 1$

(4) $\alpha y_1 + \beta y_2$ is a solution if $\alpha - \beta = 1$

113. A right circular cylinder with radius $R$ and height $H$ contains a liquid which evaporates at a rate proportional to its surface area in contact with air (proportionality constant $t = k > 0$). If $T$ is time after which cylinder will be empty, then

(1) $T$ is dependent on $R$ (2) $T$ is independent on $R$

(3) $T$ is dependent on $H$ (4) $T$ is dependent on $K$

114. The normal to the curve $y = f(x)$ passing through the point (1, 1) is $a(y - 1) + (x - 1) = 0$. If the slope of tangent at any point on the curve is proportional to the ordinate of that point, then

(1) Equation of curve is $y = e^{a(x-1)}$

(2) Equation of curve is $y = e^{ax-1}$

(3) Area bounded by the $y$-axis, the curve and the normal to the curve at $P$ is $\left|\frac{1}{a}\left(a - \frac{1}{2} + e^{-a}\right)\right|$

(4) None of these

The average birth rate and average death rate of mosquito in a certian part of India is given by $B(t) = 2t^2 - t + 1$, $D(t) = -t^2 + 3t + 4$ respectively; where $t$ = time in year. It is given that the population is $p_0 = 100$ at $t = 0$ (starting point from where the measurement is taken up) and population at any time $t$ is given by, $P(t) = \int_0^t \{B(t) - D(t)\}\,dt + P_0$ and total population at any time $t$ is divided into three groups as group of larva $A(t)$, group of youns $B(t)$, group of olds $C(t)$; where $A(t) = t^3 - 9t + 10$, $B(t) = -t^2 + 2t + 40$, $C(t) = -t^2 + 4t + 50$, at any time $t$. Negative population means absence of population. All population functions attain integer values at any time $t$. If at any time population is non-integer then it takes the nears integer value and mid-value takes higher integer (example : 7.5 is equal to 8). Calculation should be based on single digit after decimal.

115. What is the total population when populatin of group of young's is maxima?

(1) 104 (2) 101

(3) 98 (4) 96

116. What is the population of Larva when populatin is minimum ?

(1) 10 (2) 5

(3) 3 (4) Disappear

117. What is the population of 'Group of old' at the time when the population of 'Group of young's' started disappearing ?

(1) 32 (2) 25

(3) 34 (4) 43

118. After how long the group of larva started recovering their population? Choose appropriate option

(1) In one year (approximate) (2) Two year (approximate)

(3) In half year (approximate) (4) 1/4 year (approximate)

119. At the moment group of olds started disappearing, what is the percentage of Larva in total populations?

(1) 0% (2) 45%

(3) 79% (4) 100%

120. Match the properties of the curves given in Column I with the corresponding curve(s) given the column II

| **Column I** | **Column II** |
|---|---|
| (1) A curve passing through (2, 3) having the property that the length of the radius vector of any of its point *P* is equal to the length of the tangent drawn at this point, can be | (a) Straight line |
| (2) A curve passing through (1, 1) having the property that any tangent intersects the *y*-axis at the point which is equidistant from the point of tangency and the origin, can be | (b) Circle |
| (3) A curve passing through (1, 0) for which the length of normal is equal to the radius vector, can be | (c) Parabola |
| (4) A curve passes through the point (2, 1) and having the property that the segment of any of its tangent between the point of tangency and the *x*-axis is bisected by the *y*-axis, can be | (d) Hyperbola |

121. A curve *C* has the property that if the tangent drawn at any point *P* on *C* meets the co-ordinates axes at *A* and *B*, then *P* is the mid-poin of *AB*. The curve passes through the point (1, 1). Determining the equation of the curve, find $f\left(\frac{1}{5}\right)$.

122. A solution of the differential equatin $\left(\frac{dy}{dx}\right)^2 - x\frac{dy}{dx} + y = 0$ is $y = kx - \lambda$, then find the greatest value of $-24(k + \lambda)$

123. A right circular cone with radius *R* and height *H* contians a liquid which evaporates at a rate proportinal to its surface area in contact with air (proportionality constant $t = k > 0$). Find the time after which the cone is empty, when $H = 16k$.

124. A cuve passes through (1, 1) and any tangent at a point *P* on the curve is such that it intersect *x* and *y*-axis at *A* and *B* respectively. If *PA* : *PB* = 3 : 1, then finding the differential equation of the curve and solving, it is found that curve passes through the point $\left(\frac{1}{8}, \lambda\right)$; then find $\lambda$.

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